Architecture and Design Document

Software Design II - Universidad del Norte | Project 5 / RFP DH-05 - Internship and Practicum Management System

Evaluation 2 - Architecture: justifying the solution structure | Team: Sebastián Mercado - Cher Jiménez - Juan Arbeláez

Evaluation 1 answered 'do we understand the problem?'. This document answers the central question of Evaluation 2: can the team justify how the system should be structured? It does not start from technology. It starts from the evidence of Evaluation 1, turns the quality requirements into measurable scenarios, selects the few facts that drive the structure, compares alternatives, records the decisions, and only then shows the views, the data, the interactions and their failures, the contracts, the security model, the validation with its honest status, and the work that remains.

Every claim carries one of four labels. DESIGNED: decided and documented, no code. IMPLEMENTED: code exists. VERIFIED: code exists and an automated test that we ran exercises it. NOT EXECUTED: planned and not run, with the reason. Two qualifiers appear in the test plan: EXECUTED marks a check that was run but does not verify a feature by itself (the mutation check), and PARTIAL marks a test that verifies only part of what it targets. A diagram or a contract specifies behavior; it does not prove that the behavior exists.

Companion artifact	What it holds
Architecture Decision Records - DH-05	ADR-000 to ADR-010 with status, date, owner, context, decision, alternatives, consequences, verification, reconsideration triggers and trace links.
Artifacts architecture drivers and decisions.xlsx	Every table of this document as a sheet (27 sheets), in the format of the Evaluation 1 workbooks.
dh05-internship-system (repository)	README as evidence index; docs/evaluation-2 with views, ADRs, design checks, test plan, validation evidence and project status; the implemented slice and its tests.

Contents

1. Purpose and agreed scope	11. Deployment and operations
2. Significant requirements, rules and assumptions	12. Validation: test strategy, evidence and status
3. Quality attribute scenarios	13. Risks, backlog and plan
4. Architecture drivers and tensions	14. Traceability
5. Alternatives and decisions	15. What is not decided yet
6. Architecture views	16. How this evaluation builds on Evaluation 1

7. Domain, data, state and transactions	17. Evidence index
8. Critical interactions and failure behavior	18. Glossary
9. Interfaces and user-facing behavior	19. References
10. Security and privacy	

1. Purpose and agreed scope

Problem

Horizon University coordinates 600 to 900 placements a year with e-mail, spreadsheets, forms and shared folders. Nobody can say reliably in which state a placement is, a student can start before every institutional condition is met, host organizations see more student data than they need, and a placement can be closed while the academic evidence is still missing (RFP sections 1 and 2; Evaluation 1).

Desired outcome. One controlled lifecycle from opportunity to closure, gates that cannot be bypassed at activation and closure, access that follows the relationship with the placement, and a history of every decision that nobody can rewrite. Not a nicer spreadsheet.

What we take from Evaluation 1

Evaluation 1 artifact	How Evaluation 2 uses it
Quality Requirements QR-01 to QR-05	Source of the quality scenarios and of four of the five drivers.
Risk Register R-001 to R-010	Risks that became drivers: rules changing (R-001), early start (R-002), over-exposure (R-003, R-004), failing services (R-005), copying the export as schema (R-008).
Clarification and Assumption logs	Open questions the design must survive (the seven of section 15: CL-001, CL-004, CL-008, CL-016, CL-017, CL-018, CL-020) and the assumptions about authoritative services (ASM-002, ASM-003, ASM-009).
Process Analysis OB-01 to OB-12, 15 stages	The stages give the module boundaries; OB-07, OB-11 and OB-12 give the gates and the history.
Capabilities CAP-01 to CAP-06 and stories US-001 to US-030	The significant requirements of section 2; every story has a home module (section 14).
Success Criteria SC-01 to SC-12	The verification column of the traceability and the acceptance tests of the slice.

Evaluation 1 is the input of this evaluation and its artifacts stay as they were delivered. While designing, we read the data pack and the starter kit in more detail and recorded the facts that refine what we now build on (evidence updates EU-01 to EU-05):

ID	What the evidence says	How we use it
EU-01	data/reference/placement_types.csv defines ten active types (ACADEMIC_INTERNSHIP, PROFESSIONAL_PRACTICUM, COOPERATIVE_ EDUCATION, RESEARCH_INTERNSHIP, COMMUNITY_ SERVICE, TEACHING_PRACTICUM, CLINICAL_FIELD,	For the architecture we take the file as the source: ten defined types, five used by opportunities, three by placements. The Evaluation 1 artifacts stay as delivered. The question behind CL-001 is unchanged and still open (which types enter the first release is a

ID	What the evidence says	How we use it
	INTERNAL_UNIVERSITY, INTERNATIONAL, REMOTE_ INTERNSHIP). The records use fewer: internship_ opportunities.csv uses five of them (academic internship, professional practicum, community service, research internship, remote internship) and placements.csv uses three.	client decision), and the architecture does not depend on the number: placement types are reference data and rules are configured per type (ADR-004).
EU-02	Two PROFESSIONAL_PRACTICUM placements have different checklists: PLC-2001 has ACADEMIC_ APPROVAL, VALID_AGREEMENT and WORK_PLAN; PLC-2003 has VALID_AGREEMENT and INSURANCE_ REFERENCE.	The inconsistency confirms that the checklist cannot be fixed in code. The initial configuration uses the union of the observed conditions and is marked as pending CL-020.
EU-03	students_current_export.csv uses the legacy code SWE for HU-STU-5006, while academic_programs.csv lists it as an alias of BSC-SWE. placements.csv stores 'Blocked' as a status, although blocking is a consequence of the readiness checklist.	Program codes are normalized through known_aliases during import, and 'Blocked' is not a lifecycle state: it is derived from the readiness gate. Both transformations are written in data/generated/transformation_log.md.
EU-04	The starter reads data/reference/mock_users.json and exposes /mock/institutional/students, but the DH-05 pack ships mock_identities.json and a contract for /mock/student-information/students/{id}. The starter has no Employer Directory mock.	We keep the starter and write adapters for the DH-05 contracts, including a new Employer Directory mock with its three failure responses (ADR-000, ADR-007).
EU-05	mock_identities.json contains no workplace-supervisor identities (no SUP-EXT users), so the API answers 401 for them before the access policy is reached.	QS-02 is verified at policy level (test_qs02_workplace_ supervisor_is_related_only_to_their_own_placement) and through the API with an unrelated student (test_ qs02_unrelated_student_gets_404_and_denial_is_ audited). The end-to-end case with a real supervisor identity waits for the identity service or a fixture identity (BL-06).

Scope

In scope for the product:

- Opportunity registration and publication review (stages 1-3).
- Applications, eligibility decision, employer decision and academic approval as three separate results (stages 4-7).
- Placement lifecycle, work plan versions, supervisor assignment, readiness checklist, activation gate (stages 7-12).
- Hours in three levels, incidents, evaluations, closure gate (stages 13-15).
- Role and relationship access with audience projections, append-only audit history, notifications through an outbox, operational reports.

Out of scope:

- Payroll, stipends or any real financial transaction (RFP section 10).
- Automatic candidate selection or eligibility decided by artificial intelligence (RFP section 10).
- Real integrations with the identity, student-information, employer-directory and e-mail systems: they are simulated (RFP section 7).
- Storing the documents themselves: DH-05 stores evidence references only.
- Web client technology and screens: decided in Evaluation 3 because no driver depends on them.

Implemented in this evaluation as a foundation, not as the product: the activation gate (US-014, US-016) with access, audit and outbox. The other modules are designed and appear in the structure; they are built in Evaluation 3.

Stakeholders become concerns and design consequences

Stakeholder	Concern	Design consequence
Student (STK-01)	Know what is missing and when; see only feedback meant for them	Readiness projection per audience; restricted fields never leave M-07 (ADR-005).
Host organization (STK-02)	Review candidates without bureaucracy	Employer projection of an application; no academic profile (QS-01).
Workplace supervisor (STK-03)	Confirm hours and evaluate	Relationship-based access to assigned placements only (QS-02).
Faculty supervisor (STK-04)	See progress and evidence before the signoff	Closure gate owned by M-03 with evidence from M-05 (BR-06).
Academic program coordinator (STK-05)	Rules applied the same way for the whole program	Versioned rules evaluated by Strategy; the version is stored in each decision (ADR-004).
Experiential-learning coordinator (STK-06)	Never activate a placement that is not ready	Readiness gate in one transaction with a conditional update (ADR-003, ADR-010).
Legal representative (STK-07)	No activation with an expired agreement	Agreement verified in the Employer Directory at activation; fail closed (ADR-007).
IT Services (STK-08)	Maintainable, controlled integrations, small team	Modular monolith, ports and adapters, architecture test in the test suite (ADR-001, ADR-007).
Office director (STK-09)	One process, visible risk	Audit history and at-risk report (M-08, M-10).
Data privacy officer (STK-10)	Minimum data to external users	Deny by default while CL-018 is open; threat model T-02..T-04.

2. Significant requirements, rules and assumptions

Vocabulary used in every view

Term	Meaning
Opportunity	A role posted by a host organization and reviewed before students see it.
Application	A student's request for one opportunity, carrying three independent results.
Placement	The internship itself, created when a student is selected; the aggregate root of the lifecycle.
Readiness condition	One row of the checklist that must be COMPLETE and not expired before activation.
Readiness gate	The evaluation of every required condition for a placement under one rule version.
Activation	The committed change of a placement to ACTIVE. It is a decision, not the first working day.
Start date	The day the student really begins; recorded after activation.
Rule version	The identifier of the rule configuration used for one evaluation (readiness-v1).
Audit entry	An append-only record of one decision or denial with actor, values, reason and correlation id.
Outbox message	A notification written with the decision and delivered later by the worker.
Correlation id	One identifier per command, copied to its audit entry, event and messages.

Four distinctions that the whole design protects:

- Selected by the employer is not academically approved.
- Ready (the gate passes) is not active (the command committed).
- Active is not started (the start date is recorded later).
- Activated is not notified (delivery is eventual).

Significant requirements

We do not copy the thirty stories. These are the Evaluation 1 capabilities and cross-cutting stories that have architectural consequences, with the status of each one today.

ID	Requirement	Stories	Architectural significance	Status
CAP-01	Controlled publication with four review results	US-001..US-004	Opportunity lifecycle owned by M-01; review result audited.	DESIGNED
CAP-02	Eligibility checked at application time	US-005..US-008	Versioned rules (M-06) plus Student Information port with fail-soft policy (QS-05).	DESIGNED
CAP-03	Three approvals kept apart	US-009..US-011	Three separate fields and events; none implies another (BR-05).	DESIGNED
CAP-04	Readiness gate before	US-012..US-016	Gate + transition table + conditional update in	VERIFIED (slice)

ID	Requirement	Stories	Architectural significance	Status
	activation		one transaction (AD-01).	
CAP-05	Verified hours and evidence	US-017..US-021	Three levels of hours in M-05; dispute role configurable (CL-017).	DESIGNED
CAP-06	Evidence-based closure and history	US-022..US-026	Closure gate (BR-06) and append-only audit (ADR-006).	Audit VERIFIED; closure DESIGNED
US-027, US-028	Identity and relationship-based access	US-027, US-028	One access module, audience projections, deny by default (ADR-005).	VERIFIED (slice)
US-030	Notifications that never undo a decision	US-030	Transactional outbox and worker (ADR-008).	VERIFIED

Business rules and invariants that the design must enforce

ID	Rule	Source	Enforced in	Verification
BR-01	A placement becomes ACTIVE only when every required readiness condition is COMPLETE and not expired, the agreement is valid in the Employer Directory, and both supervisors are assigned.	RFP section 7; OB-07; ASM-006; R-002	M-03 + M-04 (ADR-003)	test_sc01_*, test_ sc02_*, test_ready_ placement_is_ activated_*, tests/unit/ test_readiness_rules.py
BR-02	If the agreement cannot be verified because the directory fails, the activation is refused (fail closed).	ASM-003; R-005; QS-08	M-04 AgreementValid InDirectory (ADR-007)	test_directory_down_ means_activation_ refused, test_directory_ unavailable_keeps_gate_ closed
BR-03	Activation and the real start date are different facts: the start date is recorded after activation and is not a status.	OB-07	M-03 Placement. register_start (ADR-003)	test_start_date_only_ after_activation
BR-04	'Blocked' is not a lifecycle state; it is the result of evaluating the gate.	EU-03; R-008	M-03 transition table, seed STATUS_MAP	test_active_cannot_be_ reached_by_other_ commands; transformation_log.md
BR-05	Eligibility, employer decision and academic approval are three separate results; none implies another.	OB-05; CL-003	M-02 (designed)	NOT EXECUTED - module planned for Evaluation 3
BR-06	A placement is completed only when the closure gate passes: accepted hours reach the required total, final evaluations are submitted without hold and the academic signoff exists.	OB-11; R-007; DH05-SC-05	M-03 + M-05 (ADR-003)	NOT EXECUTED - designed; closure gate is Evaluation 3
BR-07	Every state-changing command and every denied access writes an audit entry; entries are never updated or deleted.	QR-03; OB-12; US-026	M-08 (ADR-006)	test_audit_history_can_ only_be_appended_ and_read, audit assertions in activation tests
BR-08	Every eligibility or readiness decision stores the	QR-05; F-24; QS-	M-06 (ADR-004)	test_rule_change_is_

ID	Rule	Source	Enforced in	Verification
	version of the rules used; changing a rule never changes a past decision.	07		configuration_only, test_qs04_audit_reason_records_the_rule_version_used
BR-09	Restricted evaluation comments and RESTRICTED/INTERNAL incidents are visible only to authorized roles; while CL-018 is open, access is denied by default.	QR-02; CL-018; R-004	M-07 (ADR-005)	test_host_organization_sees_readiness_without_restricted_evidence (evidence references); evaluation/incident projections NOT EXECUTED (M-05 planned)
BR-10	A user reaches a placement only through a relationship with it; without one, the answer is 'not found'.	QR-02; ASM-007	M-07 (ADR-005)	test_unrelated_user_gets_not_found, test_other_organization_cannot_see_the_placement, test_qs02_*
BR-11	A failed notification never cancels or reverts an approval, an activation or a closure.	QR-04; ASM-009	M-09 outbox (ADR-008)	test_failed_delivery_keeps_activation_and_retries
BR-12	Two commands on the same placement cannot both succeed from the same starting status.	Design rule found in review (QS-09)	M-03 repository conditional update (ADR-010)	test_two_concurrent_activations_only_one_commits

Constraints

ID	Constraint	Source
C-01	Team of three students and one semester.	RFP section 7; course
C-02	Only synthetic data; integrations with identity, student, document and academic systems may be simulated.	RFP section 7; ASM-011
C-03	The university has limited technical personnel and prefers maintainable solutions.	RFP section 7; STK-08
C-04	The solution must operate independently of the other student projects.	RFP section 7
C-05	No payroll or real financial transactions; no automatic candidate selection with artificial intelligence.	RFP sections 7 and 10
C-06	A technical starter kit (Python, FastAPI, SQLAlchemy, SQLite) is supplied and may be replaced if the decision is documented.	horizon-common-starter README

Design assumptions

Evaluation 1 recorded twelve assumptions about the problem. These six are about the design; each says how we would find out that it is false and what would change. None is client-confirmed.

ID	Assumption	Source	How to validate	If false
DA-01	Load stays between 600 and 900 placements per year with 8 to 12 office users.	RFP section 8; F-22	Ask the office for last year's numbers (CL-005 follow-up).	ADR-001 revisit triggers: measure and reconsider the structure.
DA-02	One institution, one time zone; all timestamps are stored in UTC.	Starter kit convention	Confirm no campus works in another country.	Add a time-zone attribute per campus; no structural change.
DA-03	SQLite is enough for local development and the demo; PostgreSQL is the shared environment.	Starter docker-compose	Run the same tests against PostgreSQL (NOT EXECUTED, see TP-14).	Fix SQL differences; the repository isolates them.
DA-04	The mock identity header is a development mechanism, never production security.	Starter technical assumptions	Decide with IT Services the real authentication before any real pilot.	Only IdentityAdapter changes (ADR-007).
DA-05	One experiential-learning coordinator role is allowed to activate placements; other roles are not.	US-016; CL-008 open	Confirm with the office which staff roles may activate.	Change one entry of the access policy (M-07).
DA-06	The Employer Directory answers the agreement question for the organization at the moment of activation; DH-05 does not cache the answer between requests.	ASM-003	Contract test against the real service when it exists.	Add a snapshot with timestamp (already designed for verification evidence).

One synthetic scenario used throughout

Every view, sequence, test and screen uses the same records, so the reader can follow one placement across the whole document.

Record	What it holds	Where it is used
PLC-2002	Selected by ORG-001, academic approval READY-004 and work plan READY-006 pending	Refused activation (SEQ-02, DH05-SC-01)
PLC-2003	Agreement READY-007 expired, insurance READY-008 missing; ORG-005 agreement EXPIRED in the directory	Refused activation (DH05-SC-02)
PLC-9001	Synthetic test placement (not in the data pack): ORG-001, all conditions complete	Successful activation, concurrency, rollback (SEQ-01, SEQ-04)
PLC-2001	Already ACTIVE in the export, student HU-STU-5003. Its readiness read shows not ready under readiness-v1 (INSURANCE_REFERENCE not registered): a consequence of EU-02, not a defect. Its activated_at (2027-01-20) is later than the fixed test date (2026-09-	Seed exception of QS-08; unrelated-user tests

Record	What it holds	Where it is used
	24) because the synthetic pack describes a future period; we load it as exported and no rule uses that date	
EXP-COORD-001	Experiential-learning coordinator (staff role)	The only role allowed to activate
HU-STU-5001 / HU-STU-5003	Students of PLC-2002 / PLC-2001	Denied activation; not-found for unrelated placement
ORG-C-001	Contact of ORG-001	Host-organization projection; 404 for PLC-2003
Fixed date	2026-09-24 in tests	Agreement validity does not change with the calendar

Use case specification of the critical path

Use case	UC-ACT Activate a placement (US-014, US-016)
Primary actor	Experiential-learning coordinator (staff role EXPERIENTIAL_LEARNING_COORDINATOR).
Trigger	The coordinator selects Activate on a placement in PENDING_ACTIVATION.
Preconditions	Valid identity; the placement exists; the actor's staff role allows the action.
Main flow	Resolve identity, check access, read the placement, evaluate the gate under the current rule version, transition to ACTIVE, save with a conditional update, write the audit entry, publish PlacementActivated (outbox), commit, return status, rule version and correlation id.
Success guarantee	One committed ACTIVE status with activated_at, exactly one PLACEMENT_ACTIVATED audit entry and three outbox messages, all with the same correlation id.
Failure guarantee	No partial state: the placement stays PENDING_ACTIVATION, no success audit entry, no messages.
Alternatives	Gate blocks (409 ACTIVATION_BLOCKED, audited); directory down (blocks, fail closed); unrelated user (404, audited); related but not allowed (403, audited); concurrent command first (409 STATE_CHANGED); already active (409 ILLEGAL_TRANSITION); identity disabled (503).
Visible result	The office sees ACTIVE immediately; notifications may still be pending.
Status	IMPLEMENTED and VERIFIED by TP-04 and TP-05.

3. Quality attribute scenarios

A quality word is not useful for design until it is measurable. Each scenario follows the six-part form (source, stimulus, environment, artifact, response, measure) and uses a real record of the data pack or a documented failure of the simulated services, so that it can become a test. The

measures are targets for the system we build, not measurements of the current process. QS-09 was added after reviewing the first slice.

QS-01 - Privacy (QR-01)

Source	Host-organization contact ORG-C-001 (Northbridge Digital Services), authenticated as an external user.
Stimulus	Opens the candidate INT-APP-1001 (student HU-STU-5001) to take the hiring decision.
Environment	Normal operation, employer-review stage (stage 6).
Artifact	Applications module, candidate view.
Response	The system returns only the candidate projection approved for employers: name, program, the documents of that application classified as APPLICATION (CV, motivation letter) and the application status. Academic average, credits, standing, other applications of the student and RESTRICTED documents are never included.
Measure	0 fields outside the employer projection in 100% of responses to external users. Verified by an automated test per external endpoint.
Evidence	CL-009, ASM-007, document_types.csv (sensitive_level), SC-08
Priority	Business importance H / architectural impact H

QS-02 - Relationship-based Authorization (QR-02)

Source	Workplace supervisor SUP-EXT-002 (ORG-001), authenticated and with a valid supervisor role.
Stimulus	Requests the evaluations and incidents of placement PLC-2001, which belongs to ORG-002 and is supervised by SUP-EXT-001.
Environment	Normal operation.
Artifact	Access module, policy decision for every placement query.
Response	The request is denied because the user has no relationship with PLC-2001, even though their role is correct. The response does not reveal whether the placement exists, and the attempt is written to the audit history.
Measure	100% of requests without a relationship are denied; 100% of denials are recorded with actor, resource and time.
Evidence	ASM-007, CL-018, US-028, SC-08, SC-09
Priority	Business importance H / architectural impact H

QS-03 - Field-level privacy (QR-01, QR-02)

Source	Student HU-STU-5003 (placement PLC-2001).
Stimulus	Opens the midterm evaluation EVAL-001 and the incident list of their own placement.
Environment	Normal operation, monitoring stage (stage 13).
Artifact	Monitoring & Evaluation module, evaluation and incident projections.
Response	The student sees student_visible_feedback and the rating, but never restricted_comments. Incidents with visibility RESTRICTED or INTERNAL (for example INC-001 and INC-002 in the data pack) are shown only to the roles allowed for that visibility. While CL-018 is open, restricted fields are denied to everybody except the experiential-learning office, the faculty supervisor of the placement and the author of the content (a workplace supervisor reads their own comments); deny by default.
Measure	restricted_comments absent from 100% of student and external responses; tested for every audience in the access matrix.
Evidence	evaluations.csv (MIXED visibility), placement_incidents.csv (visibility), CL-018, R-004, SC-07, SC-09
Priority	Business importance H / architectural impact M

QS-04 - Auditability (QR-03)

Source	Experiential-learning coordinator EXP-COORD-001 or an auditor of the program.
Stimulus	Investigates months later why hour entry HOUR-002 was reported as 10 hours and confirmed as 8, and which work plan version was approved for PLC-2001.
Environment	Normal operation, after the placement is closed.
Artifact	Audit History module and the versioned records (work plan, decisions, hours).
Response	The history of the placement shows every change with actor, date and time, previous value, new value and reason. WP-001 is labeled v2, so the previous approved version and v2 both remain readable; nothing was overwritten.
Measure	100% of state-changing commands write an audit entry in the same database transaction as the change. No application interface can update or delete an audit entry. The history of one placement is obtained with one query.
Evidence	OB-12, F-02, DH05-SC-03, DH05-SC-04, US-026, US-021, SC-10
Priority	Business importance H / architectural impact H

QS-05 - Reliability (integration failure) (QR-04)

Source	Simulated Student Information Service.

Stimulus	Answers SIS-503 SERVICE_UNAVAILABLE (or SIS-409 STALE_ACADEMIC_PROFILE) while the eligibility of a new application is checked.
Environment	Degraded operation, peak at the beginning of the academic period (RFP section 8).
Artifact	Applications module and the Student Information adapter.
Response	The application is saved. Its eligibility stays ELIGIBILITY_PENDING with the error code recorded, and the check is retried. A 409 answer sends the case to manual review. The system never decides eligibility without a source timestamp.
Measure	0 applications lost; 0 eligibility decisions without source data and timestamp; the student sees the pending status and the reason.
Evidence	student_information_service/failure-responses.json, ASM-002, ASM-009, R-005, eligibility_reviews.csv (source_data_timestamp), SC-11
Priority	Business importance H / architectural impact M

QS-06 - Reliability (notification failure) (QR-04)

Source	Simulated notification channel.
Stimulus	Fails when the system notifies the student and the supervisors that PLC-2001 was activated.
Environment	Normal operation.
Artifact	Notifications module (outbox).
Response	The activation stays committed. The notification is kept in the outbox with its failure and is retried; the failure is visible to the office.
Measure	0 approvals, activations or closures rolled back because of a notification failure; every failed notification is retried or flagged.
Evidence	ASM-009, US-030, SC-11
Priority	Business importance M / architectural impact M

QS-07 - Maintainability (rule change) (QR-05)

Source	Academic Program Coordinator ACAD-COORD-101, through the experiential-learning office.
Stimulus	The BSc Biology program adds CONFIDENTIALITY_FORM as a readiness condition for PROFESSIONAL_PRACTICUM, and raises its minimum average, while applications are open.
Environment	Normal operation, during an open application period.
Artifact	Rule Configuration module.

Response	The change is applied as a new version of the rule configuration. New evaluations use the new version; decisions already taken keep the version they were evaluated with.
Measure	No source-code change and no redeployment; 0 past decisions modified; the rule version is visible in every eligibility decision and readiness evaluation.
Evidence	RFP section 7, F-24, CL-004, CL-020, ASM-004, R-001, EU-02
Priority	Business importance H / architectural impact H

QS-08 - Integrity of the activation gate (RFP section 7 constraint; supports QR-03, QR-04)

Source	Experiential-learning coordinator EXP-COORD-001.
Stimulus	Tries to activate PLC-2002 (academic approval and work plan pending) and PLC-2003 (agreement expired, insurance missing).
Environment	Normal operation; also when the Employer Directory Service is unavailable.
Artifact	Placements module, activation command and readiness gate.
Response	Both activations are refused and the system names the blocking conditions: READY-004 and READY-006 for PLC-2002; for PLC-2003, READY-007 (expired), READY-008 (missing), the directory's EXPIRED agreement for ORG-005 (valid until 2026-07-31), and ACADEMIC_APPROVAL and WORK_PLAN, which the configuration requires but its checklist does not contain (EU-02). If the agreement cannot be verified because the directory is down, the activation is also refused (fail closed). The attempt is recorded.
Measure	100% of activations with a pending, blocked, expired or unverifiable required condition are refused; after the initial load there is no other path (import, API, script) that sets a placement to ACTIVE. The one exception is the one-time seed, which loads historic placements as exported (PLC-2001 is already ACTIVE in the data pack) and logs it in data/generated/transformation_log.md.
Evidence	RFP section 7, ASM-006, R-002, DH05-SC-01, DH05-SC-02, SC-05
Priority	Business importance H / architectural impact H

QS-09 - Consistency under concurrent commands (RFP section 7 constraint; supports QR-03)

Source	Two experiential-learning coordinators, or one coordinator who double-clicks.
Stimulus	Both send the activation command for the same placement in PENDING_ACTIVATION at the same moment, and both read the same status before either writes.
Environment	Normal operation.
Artifact	Placements module: repository and activation command.

Response	Exactly one command writes ACTIVE. The other rolls back, receives 409 with code STATE_CHANGED and leaves no success audit entry and no notification.
Measure	For N concurrent commands on the same status, exactly 1 success audit entry and 1 set of notifications; N-1 receive STATE_CHANGED or ILLEGAL_TRANSITION. Verified by a deterministic interleaving on SQLite; true parallel run on PostgreSQL NOT EXECUTED (TP-14).
Evidence	RFP section 7, AD-01, review of the first slice (ADR-010), DH05-SC-01
Priority	Business importance M / architectural impact H

Some qualities were deliberately not turned into drivers:

- **Performance and scalability:** RFP section 8 and F-22: 600 to 900 placements per year, 8 to 12 office users and a seasonal peak at the start of each academic period. This load is small for any conventional web application, so no scenario requires special scaling decisions. We keep a revisit trigger in ADR-001.
- **24/7 high availability:** No stakeholder asked for continuous availability, and the RFP only requires that failures do not lose data (QR-04). Reliability is treated as data consistency, not as uptime.
- **Interoperability with production systems:** RFP section 7 allows simulated integrations. We protect the boundary with adapters (ADR-007) instead of designing for real institutional APIs.

4. Architecture drivers and tensions

From 30 stories, 10 stakeholders, 20 clarifications, 12 assumptions, 10 risks and 12 success criteria, five facts decide the structure. A driver is a fact that, if it changed, would change the architecture.

ID	Driver	Comes from	Architectural consequence	Decisions
AD-01	Activation and closure gates that cannot be bypassed	RFP section 7, OB-07, OB-11, ASM-006, R-002, R-007, QS-08, QS-09	Placement lifecycle changes happen only through commands of the Placements module, which evaluates the readiness gate or the closure gate in the same transaction. No other module, import or script writes the placement status.	ADR-001, ADR-002, ADR-003, ADR-010
AD-02	Access by relationship and by field	QR-01, QR-02, CL-009, CL-018, ASM-007, R-003, R-004, QS-01, QS-02, QS-03	One Access module decides every request (deny by default) using role plus relationship with the opportunity or placement, and every response goes through an audience projection that removes the fields the user may not read.	ADR-005
AD-03	Complete history that	QR-03, OB-12, F-02, US-026,	Every command writes an append-only audit	ADR-002, ADR-

ID	Driver	Comes from	Architectural consequence	Decisions
	nobody can rewrite	CL-019, QS-04, QS-09	entry in the same transaction. Work plans, eligibility decisions and hour reviews are stored as new records or versions, never overwritten.	006, ADR-010
AD-04	Rules that change by program and placement type	QR-05, RFP section 7, F-24, CL-004, CL-020, ASM-004, R-001, EU-01, EU-02, QS-07	Rules are versioned configuration evaluated by small interchangeable evaluators (Strategy). Each decision stores the rule version used.	ADR-004
AD-05	Authoritative external services that may fail	QR-04, F-17, F-18, ASM-002, ASM-003, ASM-009, R-005, service contracts, QS-05, QS-06	The core talks to ports, not to services. Adapters translate the contracts, record source timestamps and define the failure behavior of each call. Notifications leave through an outbox so that they never undo a business decision.	ADR-007, ADR-008

A trade-off is credible only when it names what gets worse. These are the tensions we resolved and the cost we accept:

Tension	Resolution	Accepted cost
Fresh agreement data vs. availability of the directory	Verify the agreement at activation and refuse when it cannot be verified	Activations stop while the directory is down (BR-02).
Immediate notification vs. reliable decision	Commit locally, deliver through the outbox	Messages arrive later and may repeat once.
Flexible rules vs. simple code	A small vocabulary of rule types configured per type and program	A rule that the vocabulary cannot express needs a new class (ADR-004 revisit).
Strong module isolation vs. one transaction	Modules in one process sharing one database, with table ownership	Isolation is enforced by tests, not by the network.
Rich history vs. minimum exposure	Audit everything, show history only to the office	External users cannot reconstruct decisions themselves.
Complete product vs. one semester	Implement the riskiest slice, design the rest and label it	Four business modules, the closure gate and the web client remain for Evaluation 3 (AR-07).

5. Alternatives and decisions

The same requirements can be built with very different structures. We compared four plausible alternatives against the drivers and the constraints of a three-person team in one semester. This is a qualitative fit analysis, not an invented benchmark.

Alternative	Main strength	Main weakness for DH-05	Verdict
Layered monolith	Simple; one transaction protects the gates.	Nothing prevents the 'controller talks to everything' decay across eleven capabilities.	Rejected
Modular monolith	Strong internal boundaries, local transactions, one pipeline.	Scales as one unit (acceptable at 600-900 placements per year); boundaries need enforcement.	Chosen (ADR-001)
Microservices	Independent deployment and scaling.	The activation gate needs data from several services: distributed transactions for a problem we do not have.	Rejected (ADR-009)
Low-code platform	Fast forms, approvals and dashboards.	Relationship and field-level access, versioned rules and custom gates are the core of DH-05 and fit low-code poorly.	Rejected (ADR-009)

The selected strategy in one sentence: one deployable backend organized in business modules with explicit interfaces, one relational database where every table belongs to one module, ports and adapters at the boundary with the institutional services, in-process domain events with an outbox for side effects, and a worker from the same codebase for delivery. It is the simplest structure that satisfies the important drivers and can evolve when the evidence changes.

Every technology presented in class went through the six questions of the decision filter:

Proposal	Problem it would solve in DH-05	Decision
Use microservices	None identified in DH-05	Reject
Use a service mesh	No service-to-service traffic exists	Reject
Use an external message broker (Kafka, RabbitMQ)	Notifications must not block or undo decisions	Reject for now (outbox instead)
Use serverless functions for reminders	Deadline reminders are occasional	Reject for now; revisit if reminders grow
Add a RAG policy assistant	Students and staff may not understand program rules	Not adopted in this release
Use adapters for external services	SIS and Employer Directory are outside our control and fail	Adopt (ADR-007)
Process notifications asynchronously	A notification failure must not cancel an activation	Adopt (ADR-008)
Configurable rules (Strategy)	Rules change by program and placement type	Adopt (ADR-004)

Decision register

ID	Decision	Drivers	Verified by
ADR-000	Adopt the Horizon common starter kit as technical baseline	C-01, C-03, C-06, AD-05	Starter tests still pass (tests/importers, test_health_endpoint.py, test_sample_endpoint.py); the DH-05 files are used by the adapters.
ADR-001	Use a modular monolith as architectural style	AD-01, AD-02, AD-03, AD-05, C-01, C-03	tests/architecture/test_module_boundaries.py fails when a module imports another one outside the allowed table; one runnable artifact (API + worker).
ADR-002	One relational database with module-owned tables and transactional commands	AD-01, AD-03	test_failure_in_the_middle_rolls_everything_back; test_two_concurrent_activations_only_one_commits. PostgreSQL run: NOT EXECUTED (TP-14).
ADR-003	Explicit placement lifecycle with gated transitions	AD-01, AD-03	tests/unit/test_placement_lifecycle.py (6); activation tests (8). Closure gate: DESIGNED only.
ADR-004	Versioned rule configuration evaluated through Strategy	AD-04, QR-05	test_rule_change_is_configuration_only; test_qs04_audit_reason_records_the_rule_version_used. Eligibility rules: DESIGNED.
ADR-005	Centralized relationship-based access with audience projections	AD-02, QR-01, QR-02	test_unrelated_user_gets_not_found; test_host_organization_sees_readiness_without_restricted_evidence; test_qs02_*. Evaluation/incident projections: DESIGNED.
ADR-006	Append-only audit history and versioned records	AD-03, QR-03	test_audit_history_can_only_be_appended_and_read; audit assertions in activation tests; test_one_correlation_id_links_audit_entry_and_notifications.
ADR-007	Ports and adapters for identity, student information and employer directory	AD-05, QR-04, ASM-002, ASM-003	tests/unit/test_reference_and_adapters.py; test_directory_down_means_activation_refused; architecture test: mocks imported only by M-11 and composition points.
ADR-008	In-process domain events with a transactional outbox for side effects	AD-05, QR-04, AD-03	tests/integration/test_outbox.py; test_qs06_worker_crash_*; test_qs06_outbox_payload_*.
ADR-009	Do not adopt microservices, service mesh, serverless, low-code or RAG in this release	AD-01 to AD-05, C-01, C-05	Not a testable decision; reviewed against the decision filter. Revisit triggers recorded in ADR-001.
ADR-010	Concurrent commands: conditional update on status and a correlation id per command	AD-01, AD-03	test_two_concurrent_activations_only_one_commits; test_one_correlation_id_links_audit_entry_and_notifications; test_refusal_and_denial_carry_machine_readable_codes. Parallel PostgreSQL run: NOT EXECUTED.

The full records, including ADR-010 (conditional update and correlation id, added after the slice review), are in 'Architecture Decision Records - DH-05'.

Patterns and principles, with what we do not claim

Choice	Where	Why it is justified	What it does not do
Strategy + Factory Method	M-06 evaluators and Rule EvaluatorFactory	Rules change by program; a new rule is configuration or one new class	It does not let anyone write arbitrary logic in the JSON file.
Adapter (ports and adapters)	M-11 adapters, ports in the shared kernel	Mock contracts and failures stay outside the core	It does not make a real integration correct; contracts must be tested against the real service.
Repository + conditional update	SqlPlacementRepository.save	Persistence hidden from the use case; lost update impossible	It does not lock rows; the loser still runs the gate before failing.
Unit of work (one session per command)	ActivatePlacement.execute	State, audit and outbox commit together	It does not make the directory call atomic with the write.
Observer (in-process events) + outbox	EventDispatcher, on_ placement_activated, deliver_pending	The use case does not know who reacts; delivery failures never undo a decision	It does not prevent one duplicate message after a crash between send and commit.
Facade per module	Public functions of each module	Modules do not reach into each other	A folder alone does not enforce it; the architecture test does.

We reviewed the whole refactoring.guru catalog and the enterprise patterns of the course. State was considered for the lifecycle and Proxy for access control; both were rejected with a written reason (Design Patterns sheet). No problem, no pattern.

6. Architecture views

One system, different questions, different views. Each view answers the question of one audience and leaves out what that audience does not need; the same names are used everywhere. All figures share one notation: the traceability ID of each element is on its tab, teal is inside DH-05, plum is people, grey is outside DH-05, dashed means designed but not built, and the saffron highlight marks the critical path of the activation gate (AD-01).

View	Audience	Question answered
V-01 Context	Sponsor, office director, IT Services	Who uses DH-05 and which institutional systems surround it?
V-02 Container	Architect and developers	Which deployable parts collaborate and where is the data?
V-03 Component	Backend team	How is the API divided into modules and which ones

View	Audience	Question answered
		may call which?
V-04 Component (zoom)	Developers of M-03 and M-04	How does the activation gate work inside the Placements module?
V-05 Deployment	Operations / IT Services	Where does the software run and which connections cross trust boundaries?
V-DEV Development	Developers	Which package may import which, and what checks it?
V-DOM, V-DATA	Developers, reviewers	Which concepts and tables exist, and who owns them?
V-TXN, V-STATUS, SEQ-01..05	Backend team, reviewers	What happens inside a command and when something fails?
V-UX-01, V-UX-02	Product owner, users	How does each role reach its work, and what does the screen show?

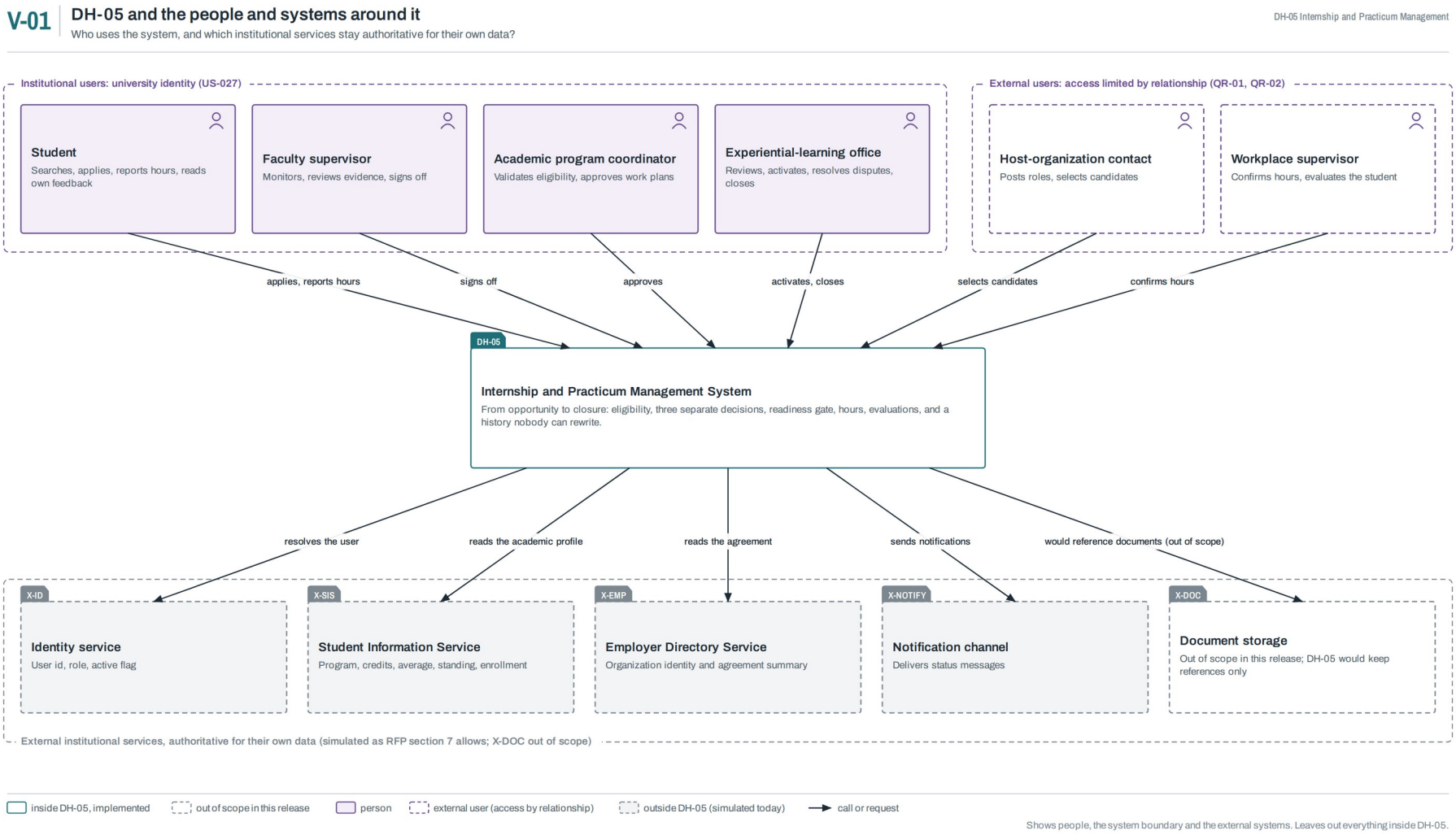


Figure 1. V-01 Context view.

The external users have their own boundary because they are the reason for AD-02 and QR-01. Four services are simulated, as the RFP allows, and they stay authoritative: DH-05 reads what it needs for each decision and never keeps the academic record or the agreement (ASM-002, ASM-003). Document storage is out of scope in this release; it is drawn because DH-05 would keep only references to documents.

Context relationship catalog

External party	DH-05 expects	DH-05 provides	Failure consequence
Identity (X-ID)	A user id, role and active flag for the caller	The user id in the header (development) or SSO later	401 if unknown or inactive; 503 if disabled. No local identities are invented.
Student Information Service (X-SIS)	Program, credits, average, standing, enrollment, with a timestamp	Read requests only	503: application kept as pending and retried; 409: manual review; 404: business error (designed for M-02).
Employer Directory (X-EMP)	Organization identity and agreement status with valid-until date	Read requests only	Any failure at activation blocks it (fail closed).
Notification channel (X-NOTIFY)	Acceptance of a message	Recipient and template; no restricted data	Message stays PENDING and is retried; FAILED after 5 attempts.
Document storage (X-DOC)	The documents themselves	Evidence references only	Out of scope in this release; the reference remains valid text.

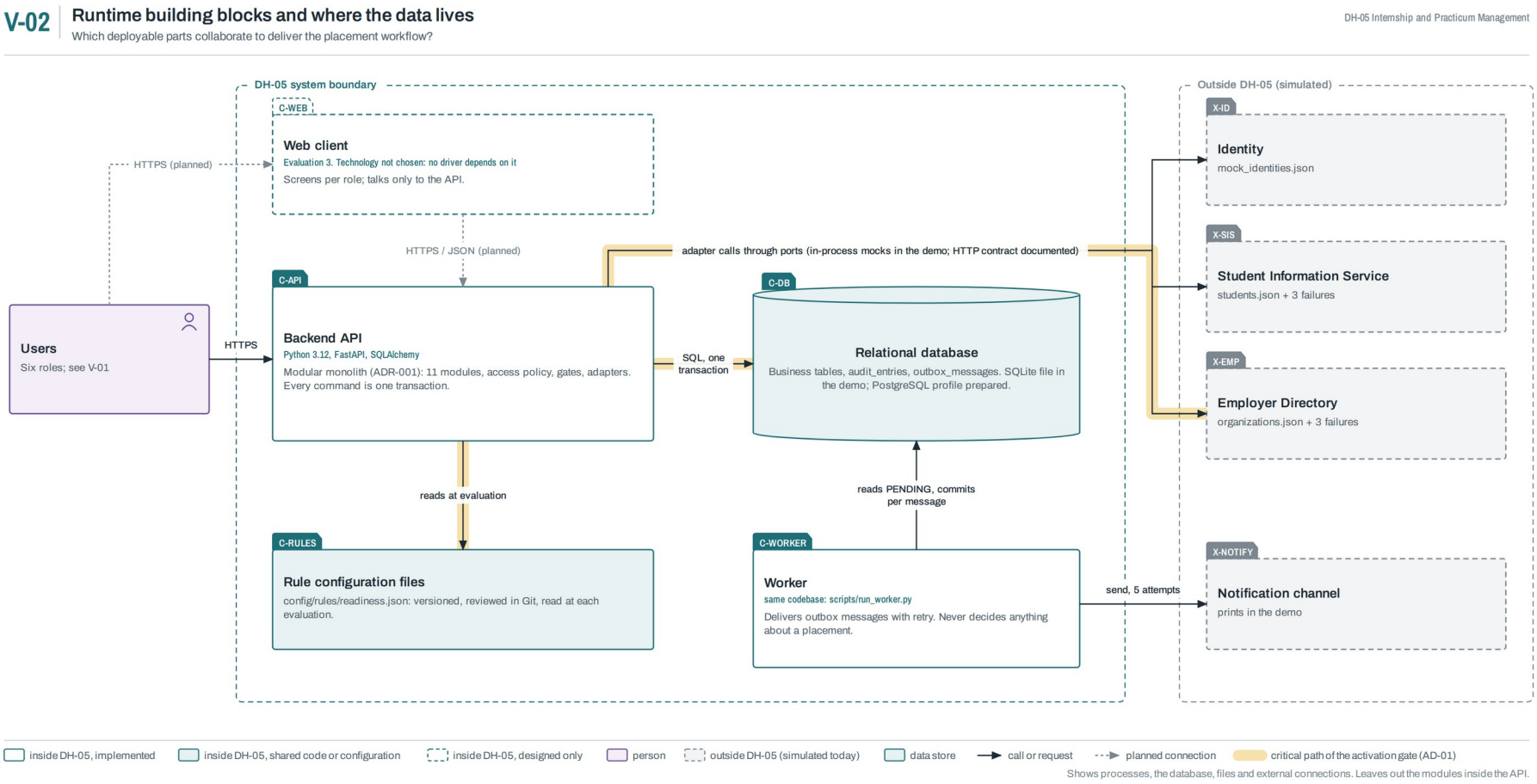


Figure 2. V-02 Container view.

The worker is the same artifact started in another mode; it exists only because notifications must never block or undo a decision (ADR-008). The database stores the state change, its audit entry and the outbox message in the same transaction (ADR-002). In the demo the external services are in-process mocks reached through the ports; the HTTP contract is documented for the real services.

Container responsibilities and non-responsibilities

Container	Owns	Does not own
C-API Backend API	Commands and reads, access decisions, gates, rules, adapters, the transaction of each command	Delivering notifications; storing documents; authoritative student or agreement data
C-WORKER Worker	Reading PENDING outbox messages, delivery, retries, marking SENT or FAILED	Deciding anything about a placement; re-evaluating the gate
C-DB Relational database	Business tables owned by modules, audit_entries, outbox_messages	Business rules; sending messages
C-RULES Rule files	The versioned rule configuration	The decisions taken with it (stored by the modules)
C-WEB Web client (planned)	Screens per role, input validation for convenience	Authorization decisions; direct database access

Modules

A module is a responsibility boundary, not a folder. For every module: the capability it owns, the data it controls, the interface it offers and what it does not own. Status: M-03, M-04, M-06, M-07, M-08, M-09 and M-11 have code; M-01, M-02, M-05 and M-10 are designed.

Module	Owns	Exposes	Does NOT own
M-01 Opportunities (stages 1-3)	Registration requests, opportunities with their criteria, review results with reason and reviewer.	register_organization, post_ opportunity, review_ opportunity, search_ published(program)	Organization identity and agreement summary (Employer Directory); eligibility rules (M-06).
M-02 Applications (stages 4-7)	Applications, eligibility decisions (rule version, per-rule result, source timestamp), employer decisions, academic approvals.	submit_application, request_ self_arranged, record_ employer_decision, record_ academic_approval, application_ status	Student academic data (Student Information Service); readiness conditions (M-04); the placement lifecycle (M-03).
M-03 Placements (stages 7, 10-12, 15)	Placements and their status transitions, work-plan versions, supervisor assignments, activation and start dates, closure result.	create_from_application, approve_work_plan, assign_ supervisors, activate, register_ start, close, placement_summary	Readiness evidence (M-04); hours and evaluations (M-05); rules (M-06).
M-04 Readiness & Documents (stages 8-9)	Readiness conditions, evidence references, verification results, validity dates.	checklist(placement), evaluate(facts, today) -> Gate Result(rule_version, blocking); register_evidence and verify_ condition are designed	The documents themselves (external document storage, simulated); the agreement master data (Employer Directory).
M-05 Monitoring &	Hour entries and reviews, progress reports, incidents with visibility,	report_hours, review_hours, resolve_dispute, report_	The placement status (M-03); who may read restricted fields

Module	Owns	Exposes	Does NOT own
Evaluation (stages 13-14)	evaluations (student-visible and restricted fields), signoff.	incident, submit_evaluation, give_signoff, completion_ evidence(placement)	(M-07).
M-06 Rule Configuration (stages 5, 8, 14)	Rule sets, versions, parameters, reference data (programs with aliases, placement types, document types).	evaluate_eligibility(profile, opportunity), readiness_ requirements(type, program), completion_requirements(type, program), current_version	The decisions themselves (stored by M-02, M-04, M-05).
M-07 Access (stages All)	Relationships (student of, contact of, supervisor of, coordinator of), access policy, projections.	actor_for(identity), audience(actor, relations), check(actor, action, relations), project_condition(audience, condition)	Identities (Identity service, simulated).
M-08 Audit History (stages All)	Audit entries.	record(...) inside the current transaction, history(resource); nothing else	Business data.
M-09 Notifications & Reminders (stages All)	Outbox messages, delivery attempts, reminder schedule.	handles domain events; deliver_ pending() in the worker	Decisions; the notification channel (simulated).
M-10 Reporting (stages All)	Read-only queries and report definitions.	placements_by(program, organization, status), at_risk()	Any writable data.
M-11 Integration Adapters (stages All)	Translation of payloads, source timestamps, failure mapping.	Implementations of IdentityPort, StudentInformationPort, EmployerDirectoryPort (the ports are defined in the shared kernel)	Authoritative student, identity or employer data.

Dependency rules:

1. A module calls another module only through the public classes listed in its 'Exposes' column (today: ReadinessGate, AccessPolicy, AuditRecorder); it never queries another module's tables. Exception by design: M-10 Reporting reads through read-only queries it owns (ADR-002). A single facade object per module is designed, not implemented.
2. The shared kernel (horizon/shared: ports, domain events) may be used by every module. Ports are defined there; only M-11 implements them and only M-11 knows the URLs and payloads of external services.
3. M-06 Rule Configuration, M-07 Access and M-08 Audit History are leaf modules: they import no other module. M-09 and M-11 also import no module (they use the shared kernel).

4. M-03 Placements may use M-04, M-07 and M-08 (implemented). M-04 may use M-06. These rules are checked by tests/architecture/test_module_boundaries.py, which reads the imports of the code.
5. M-03 Placements needs completion evidence from M-05, and M-05 needs to know if a placement is active. To avoid a cycle, M-03 will define CompletionEvidencePort and M-05 will implement it (dependency inversion). Designed, not implemented.
6. Readiness conditions completed by other modules (academic approval, approved work plan) will be updated through domain events, not by direct calls (designed).
7. Access is decided only by M-07, but the audit of a denial is written by the calling use case, because M-07 does not depend on M-08.
8. Routers (api.py) are the composition points: they may wire adapters, mocks and modules together; domain and rule code may not import frameworks or mocks.

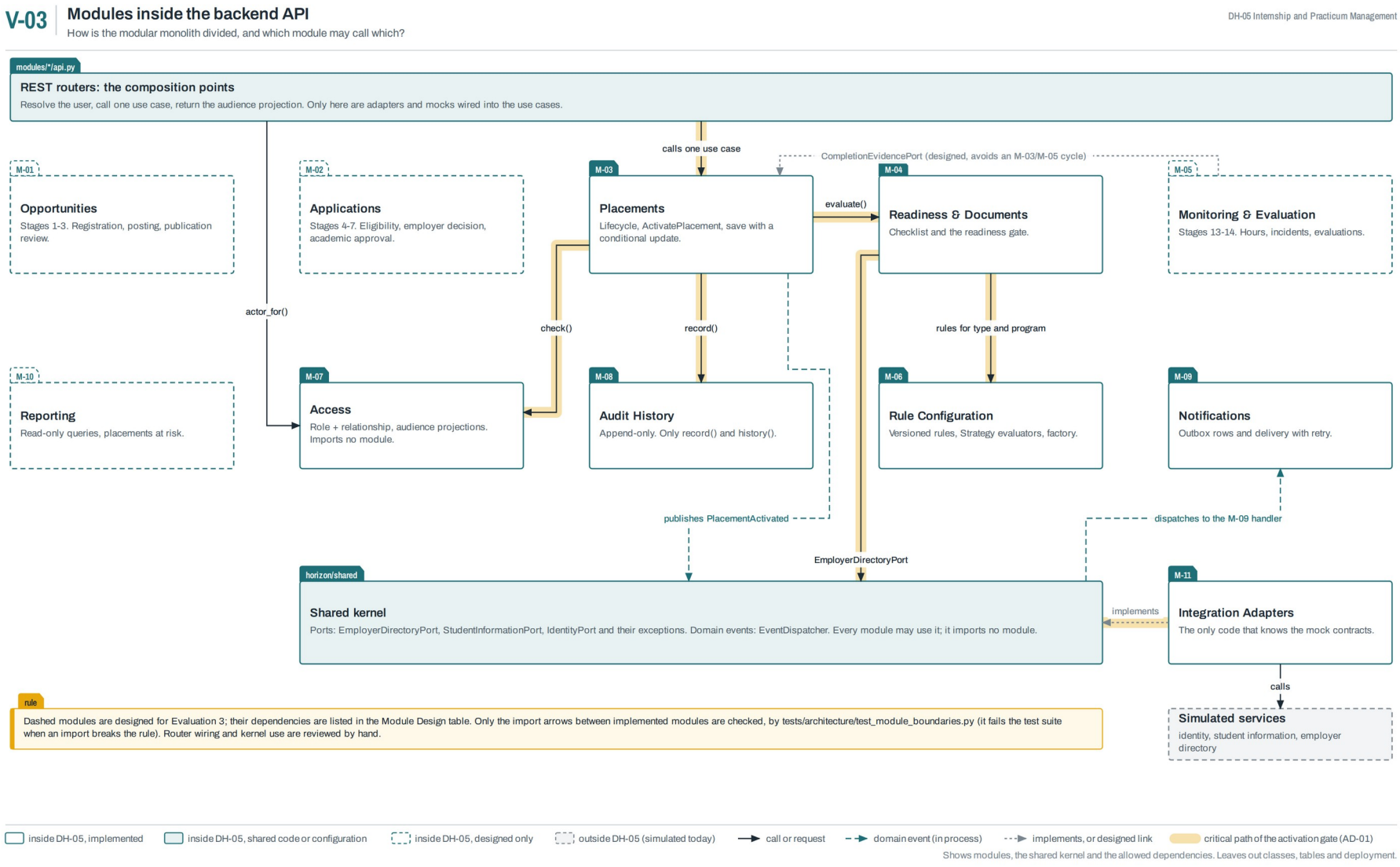


Figure 3. V-03 Component view of the backend API: modules, shared kernel and allowed dependencies.

Where does 'accepted hours must reach the required total' belong? In M-05, which owns the hours. M-03 only asks for the completion evidence through CompletionEvidencePort when it evaluates the closure gate; this avoids a cycle between Placements and Monitoring.

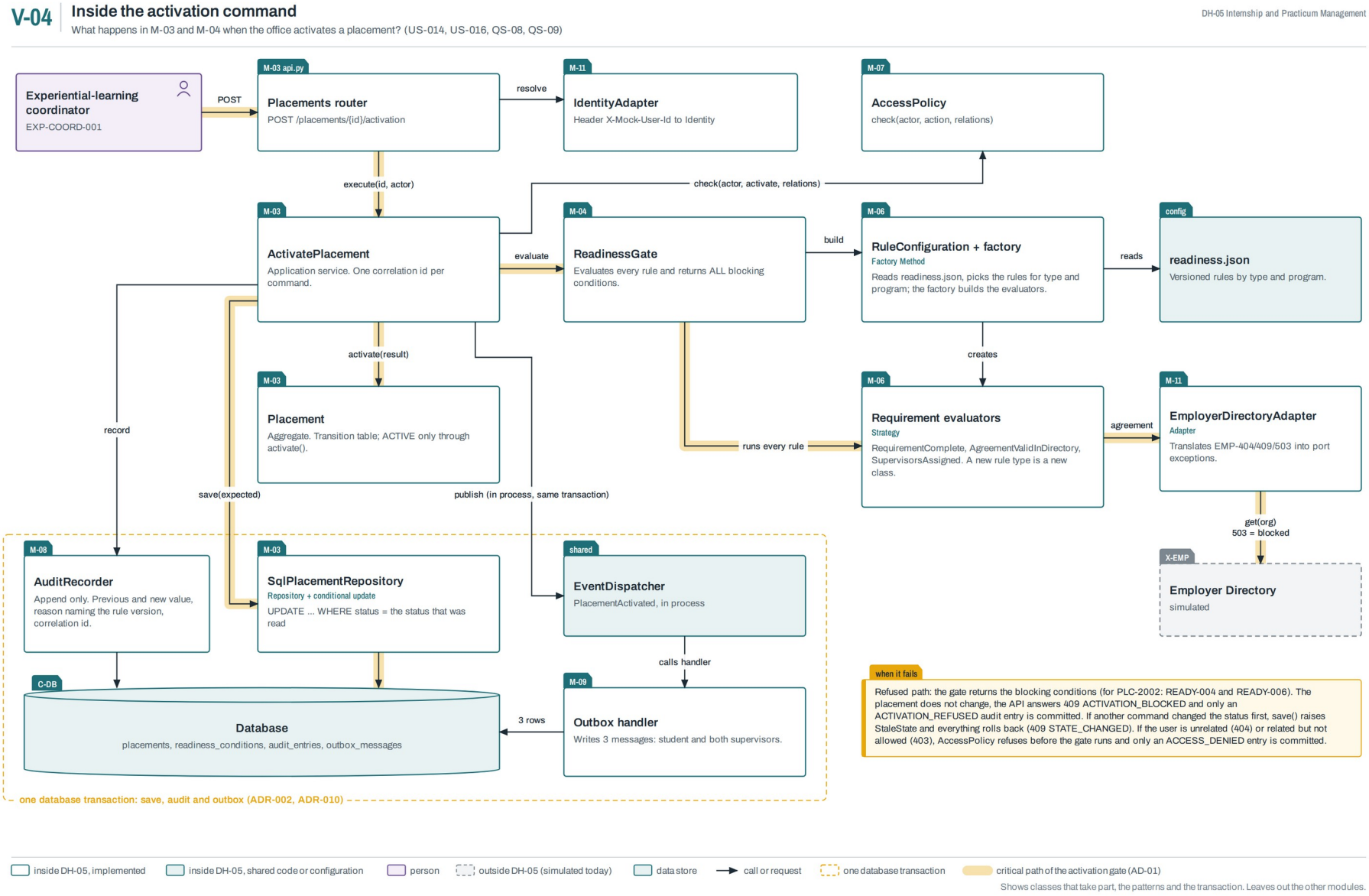


Figure 4. V-04 Component view (zoom) of the activation command.

Patterns appear only where a design problem made them necessary: Strategy and Factory Method for rules that change, Adapter at the external boundary, Repository with a conditional update for the write, Observer and outbox for side effects.

Component specification: ActivatePlacement

Component	ActivatePlacement (M-03 application service)
Purpose	Apply the activation command while keeping BR-01, BR-07, BR-11 and BR-12.
Provided operation	execute(placement_id, actor) -> {placement_id, status, rule_version, correlation_id}
Inputs	Actor (from IdentityAdapter and AccessPolicy.actor_for), placement id, today and now (injected clocks).
Dependencies	AccessPolicy (M-07), ReadinessGate (M-04), SqlPlacementRepository and Placement (M-03), AuditRecorder (M-08), EventDispatcher (shared kernel).
State owned	The placement status and activated_at.
Failure contract	PlacementNotFound, AccessDenied(reveal), ActivationBlocked(blocking, correlation_id), IllegalTransition, StaleState, any other exception rolls back.
Forbidden	Calling a mock directly; writing the status without the expected status; committing the audit entry in a separate transaction from the change.
Verification	TP-04, TP-05; mutation M1 (no conditional update) and M4 (student may activate) are detected.

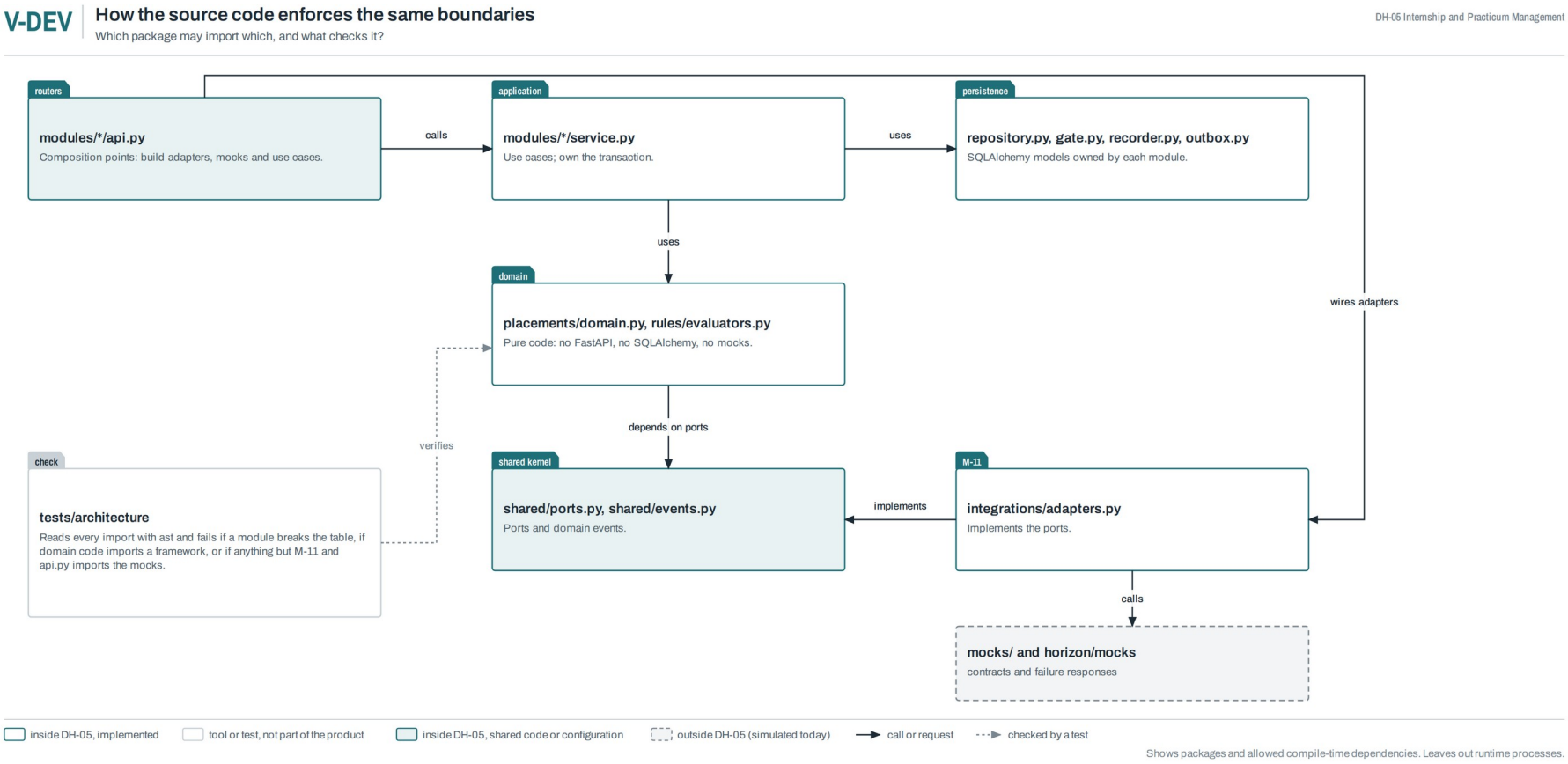


Figure 5. V-DEV Source-code organization and the test that enforces it.

Tests detect forbidden imports instead of relying on a README statement: mutation M5 (a forbidden import) makes the architecture test fail.

Cross-view consistency ledger

Element	Context	Container	Component / design	Deployment
DH-05	The system	Boundary around C-API, C-WORKER, C-DB, C-RULES	Scope of V-03 and V-04	Demo laptop or CI runner; pilot host (designed)
C-API	Hidden	Backend API	Routers, modules M-01..M-11, shared kernel	uvicorn process
C-WORKER	Hidden	Worker	deliver_pending in M-09	scripts/run_worker.py process
C-DB	Hidden	Relational database	Tables owned by M-03, M-04, M-08, M-09	SQLite file (demo); PostgreSQL (prepared)
X-ID, X-SIS, X-EMP	External systems	Outside DH-05	Ports + M-11 adapters	In-process mocks (demo); real services (pilot)
X-NOTIFY	External system	Outside DH-05	Outbox + worker	Printed by the worker (demo)

7. Domain, data, state and transactions

The model comes from the DH-05 data pack, not from the starter's generic table, and it is divided by module: every table has one owner (ADR-002). We do not copy the export blindly: 'Blocked' is not stored, program codes are normalized and decisions get a rule version.

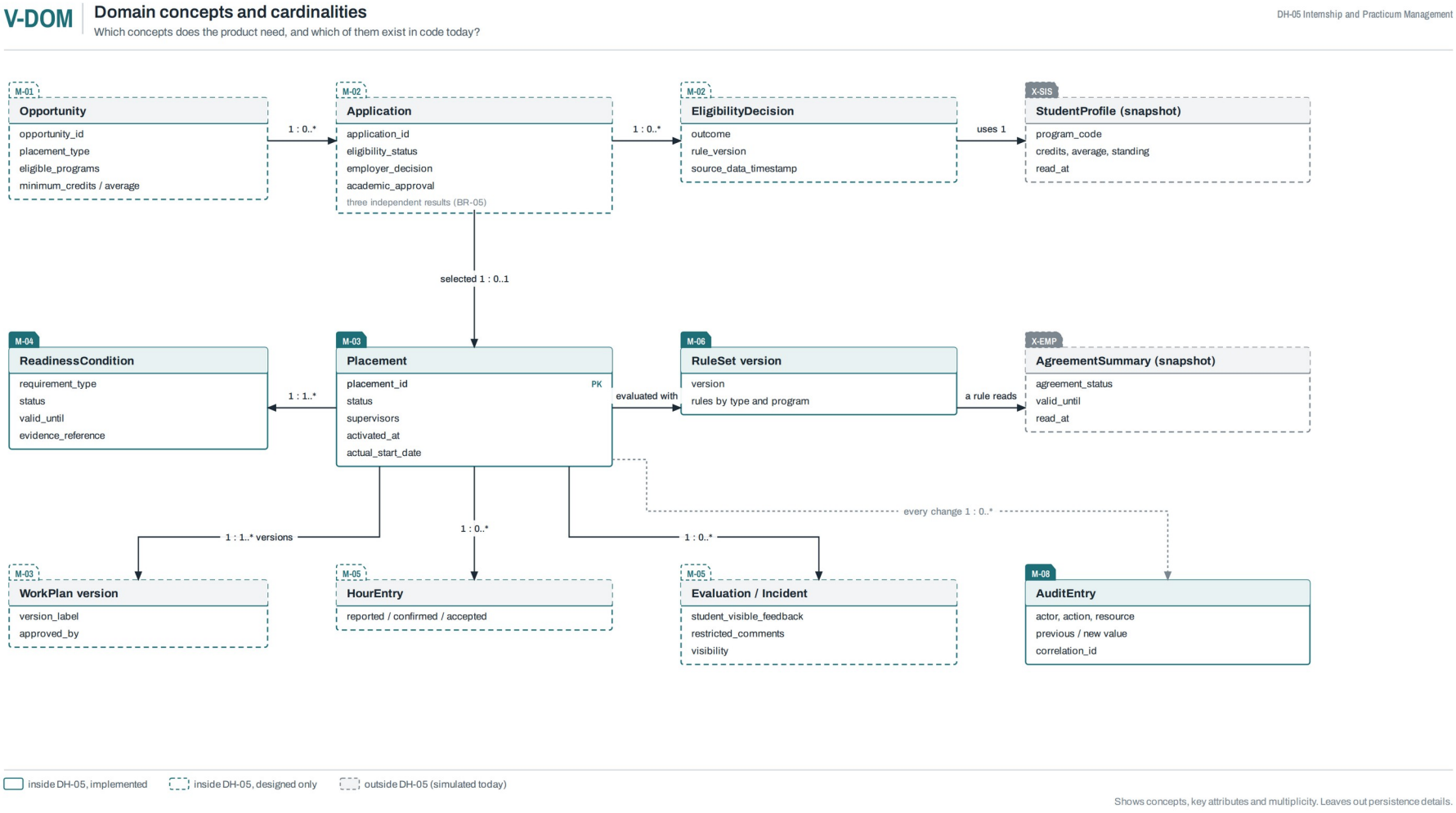


Figure 6. V-DOM Domain concepts and cardinalities. Dashed concepts are designed for Evaluation 3; grey ones are snapshots owned by external services.

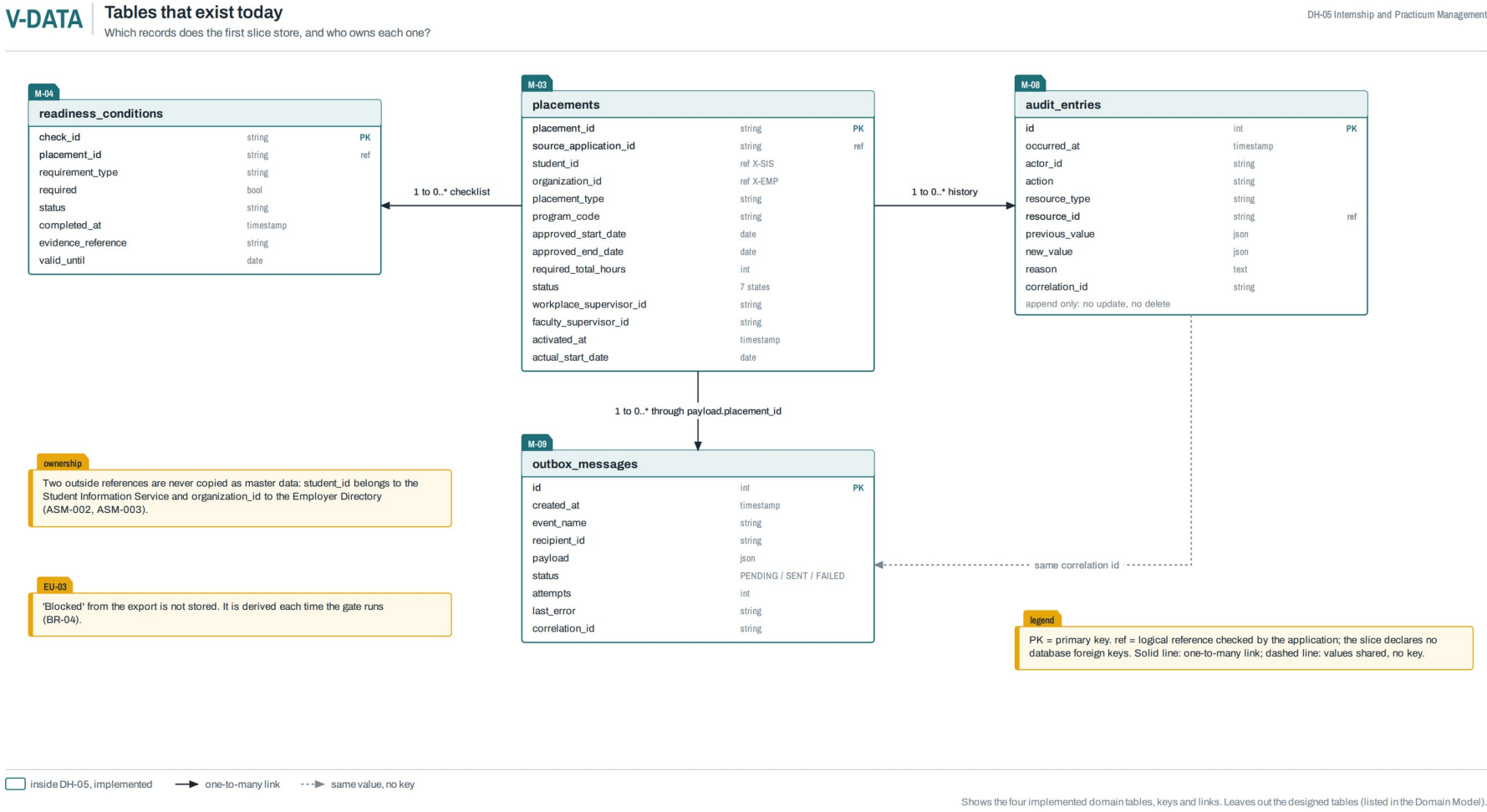


Figure 7. V-DATA The four domain tables that exist today (the starter's imported_records staging table is not drawn).

Entity	Module	Status	Key attributes	Notes
Placement	M-03	IMPLEMENTED	placement_id (PK), source_application_id, student_id, organization_id, placement_type, program_code, approved_start_date, approved_end_date, required_total_hours, status, workplace_supervisor_id, faculty_ supervisor_id, activated_at, actual_start_date	Aggregate root of the lifecycle. Source: placements.csv; 'Blocked' is not stored.
Readiness Condition	M-04	IMPLEMENTED	check_id (PK), placement_id, requirement_type, required, status, completed_at, evidence_ reference, valid_until	Checklist row. Source: readiness_checklist.csv. valid_ until is the expiry used by the gate (null for every seeded row: the export has no such column).
AuditEntry	M-08	IMPLEMENTED	id (PK), occurred_at, actor_id, action, resource_ type, resource_id, previous_value, new_value, reason, correlation_id	Append-only. Written in the same transaction as the change.
OutboxMessage	M-09	IMPLEMENTED	id (PK), created_at, event_name, recipient_id, payload, status, attempts, last_error, correlation_id	PENDING, SENT or FAILED (after 5 attempts).
RuleSet (file)	M-06	IMPLEMENTED	version, readiness rules by placement type and program	config/rules/readiness.json, lookup TYPE@PROGRAM, TYPE, default.
Opportunity	M-01	DESIGNED	opportunity_id, organization_id, title, placement_type, eligible_programs, work_ modality, dates, minimum_credits, minimum_ average, status	Source: internship_ opportunities.csv.
Application	M-02	DESIGNED	application_id, student_id, opportunity_id, application_status, eligibility_status, employer_ decision, academic_approval, current_version	Three separate results (BR-05). Source: applications.csv.
EligibilityDecision	M-02	DESIGNED	review_id, application_id, outcome, criteria_ summary, source_data_timestamp, rule_version	Source: eligibility_reviews.csv; rule_version is new.
WorkPlan (version)	M-03	DESIGNED	work_plan_id, placement_id, version_label, approved_at, approved_by, current_version	New record per version, never overwritten. Source: work_ plans.csv.
HourEntry	M-05	DESIGNED	hour_entry_id, placement_id, work_date, reported_hours, confirmed_hours, accepted_ hours, statuses	Three levels of hours. Source: hour_entries.csv.
Evaluation	M-05	DESIGNED	evaluation_id, placement_id, evaluation_type, student_visible_feedback, restricted_ comments, visibility_classification	Mixed visibility in one record. Source: evaluations.csv.
Incident	M-05	DESIGNED	incident_id, placement_id, category, severity, visibility, status	Visibility drives the projection. Source: placement_incidents. csv.
StudentProfile (snapshot)	M-11 / M-02	IMPLEMENTED (port)	student_id, program_code, enrollment_status, completed_credits, academic_average,	Not stored as master data; owned by the Student

Entity	Module	Status	Key attributes	Notes
			academic_standing, read_at	Information Service.
Agreement Summary (snapshot)	M-11 / M-04	IMPLEMENTED (port)	organization_id, active, agreement_status, agreement_valid_until, read_at	Owned by the Employer Directory. Transient value returned by the port; not persisted.

Data dictionary of the implemented tables

Field	Meaning and rule
placements.status	One of the seven lifecycle states. Today only the activate command writes it, through the transition table and conditional update; the other commands are DESIGNED to reuse the same path (ADR-010).
placements.activated_at	Instant of the activation commit (UTC). Null before activation.
placements.actual_start_date	Real first working day, recorded after activation; never a status.
placements.program_code	Normalized through academic_programs.known_aliases (SWE becomes BSC-SWE).
readiness_conditions.valid_until	Expiry of the evidence; a COMPLETE row past this date blocks. The exported checklist has no such column, so the seed leaves it null; only synthetic test rows set it.
readiness_conditions.evidence_ reference	Reference to a document kept elsewhere; null in projections for RESTRICTED requirements.
audit_entries.previous_value / new_ value	JSON snapshots of the changed fields; a refusal stores the blocking list.
audit_entries.reason	Free text. Gate decisions name the rule version ('readiness gate readiness-v1' on refusal, 'readiness gate readiness-v1 passed' on success); ACCESS_DENIED entries hold the attempted action (for example placement.activate).
audit_entries.correlation_id	Same value as the event, the outbox messages and the API response.
outbox_messages.payload	Only placement_id and template; no personal or restricted data.
outbox_messages.attempts / status	Incremented on each failed delivery; FAILED at 5.

Data ownership

Information	Authoritative source	What DH-05 keeps
Student identity and academic profile	Student Information Service	Nothing stored today; the eligibility decision (M-02, designed) will record the values it used and their read_at time

Information	Authoritative source	What DH-05 keeps
Organization identity and agreement summary	Employer Directory Service	Organization reference only; the agreement is read at each gate evaluation and not stored (a refusal records the status that blocked)
User identity and institutional role	Identity service	User id in audit entries and relationships
Opportunities and reviews	DH-05 (M-01)	Master data
Applications, eligibility, employer decision, academic approval	DH-05 (M-02)	Master data; decisions append-only
Placement, lifecycle, work-plan versions, supervisors	DH-05 (M-03)	Master data; versions append-only
Readiness conditions and evidence references	DH-05 (M-04)	Master data; documents stay in the document system (simulated reference)
Hours, progress, incidents, evaluations, signoff	DH-05 (M-05)	Master data; reviews append-only
Rule configuration and reference data	DH-05 (M-06)	Versioned configuration
Audit history	DH-05 (M-08)	Append-only

Invariants and the mechanism that enforces each one

ID	Invariant	Enforced in	Status	Verified by
INV-01	A placement status is always one of the seven lifecycle states and changes only through the transition table.	Placement._move, TRANSITIONS	VERIFIED	tests/unit/test_placement_lifecycle.py
INV-02	ACTIVE can be reached only by 'activate' with a passed gate result.	Placement.activate	VERIFIED	test_activation_needs_a_passed_gate, test_active_cannot_be_reached_by_other_commands
INV-03	A start date exists only on a placement that was activated.	Placement.register_start	VERIFIED	test_start_date_only_after_activation
INV-04	Each state change is stored only if the stored status still equals the status read (conditional update).	SqlPlacementRepository.save	VERIFIED	test_two_concurrent_activations_only_one_commits
INV-05	State change, audit entry and outbox messages of one command commit together or not at all.	ActivatePlacement.execute	VERIFIED	test_failure_in_the_middle_rolls_everything_back
INV-06	Audit entries can only be appended and read.	AuditRecorder public interface	VERIFIED	test_audit_history_can_only_be_appended_and_read
INV-07	One correlation id links the audit entry, the event and the outbox messages of a command.	new_correlation_id	VERIFIED	test_one_correlation_id_links_audit_entry_and_notifications

ID	Invariant	Enforced in	Status	Verified by
INV-08	The seed never modifies the raw data-pack files.	seed_placements	VERIFIED	test_seed_never_modifies_ the_raw_files

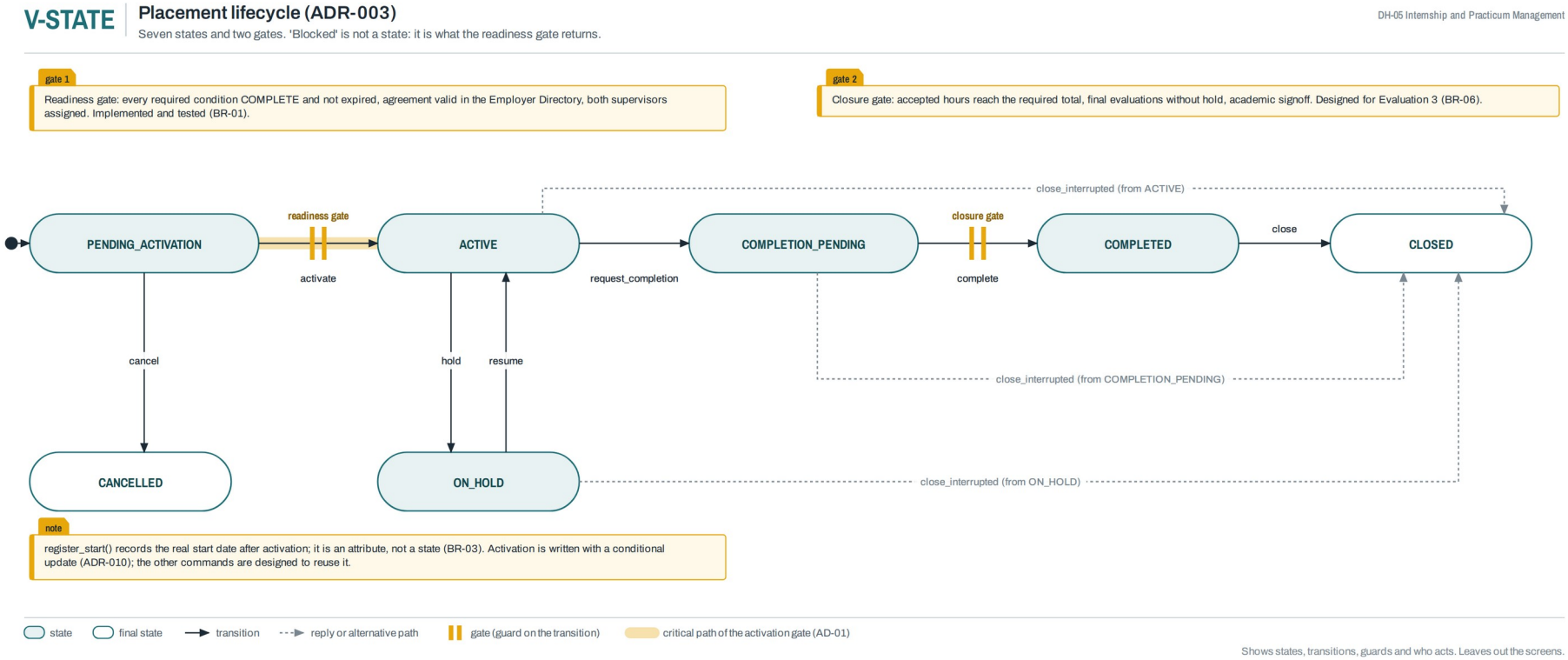


Figure 8. Placement lifecycle (ADR-003). Two transitions have gates; 'Blocked' is a result of the gate, not a state.

From	To	Command	Actor	Guard	Committed side effects	Status
PENDING_ ACTIVATION	ACTIVE	activate	Experiential-learning coordinator	Readiness gate passes; status still PENDING_ ACTIVATION at write	activated_at, audit PLACEMENT_ ACTIVATED, 3 outbox messages	VERIFIED
PENDING_ ACTIVATION	(unchanged)	activate refused	Same	Gate blocks	audit ACTIVATION_ REFUSED with blocking list	VERIFIED
PENDING_ ACTIVATION	CANCELLED	cancel	Coordinator	Reason required	audit entry	Transition table only
ACTIVE	ON_HOLD	hold	Coordinator	Reason required	audit entry	Transition table only
ON_HOLD	ACTIVE	resume	Coordinator	Reason required	audit entry	Transition table only
ACTIVE	COMPLETION_ PENDING	request_ completion	Student / coordinator	End date reached or final evidence submitted	audit entry, reminders	Transition table only
COMPLETION_ PENDING	COMPLETED	complete	Faculty supervisor	Closure gate	audit entry, notifications	DESIGNED
COMPLETED	CLOSED	close	Coordinator	Coordinator audit done	audit entry	Transition table only
ACTIVE / ON_ HOLD / COMPLETION_ PENDING	CLOSED	close_ interrupted	Coordinator	Reason and result	audit entry	Transition table only

Status in placements.csv	Lifecycle state	Note
Activation pending (PLC-2002)	PENDING_ACTIVATION	Gate shows READY-004 and READY-006 pending
Blocked (PLC-2003)	PENDING_ACTIVATION	'Blocked' is derived from the gate, not stored. For PLC-2003 the gate names READY-007 (expired), READY-008 (missing), the directory agreement EXPIRED and two required conditions not registered
ACTIVE (PLC-2001)	ACTIVE	activated_at kept (2027-01-20T08:00:00Z); actual_ start_date not present in the export, left null
Completion pending (PLC-2004)	COMPLETION_PENDING	The closure gate (DESIGNED) would show EVAL-002 overdue and EVAL-003 with hold

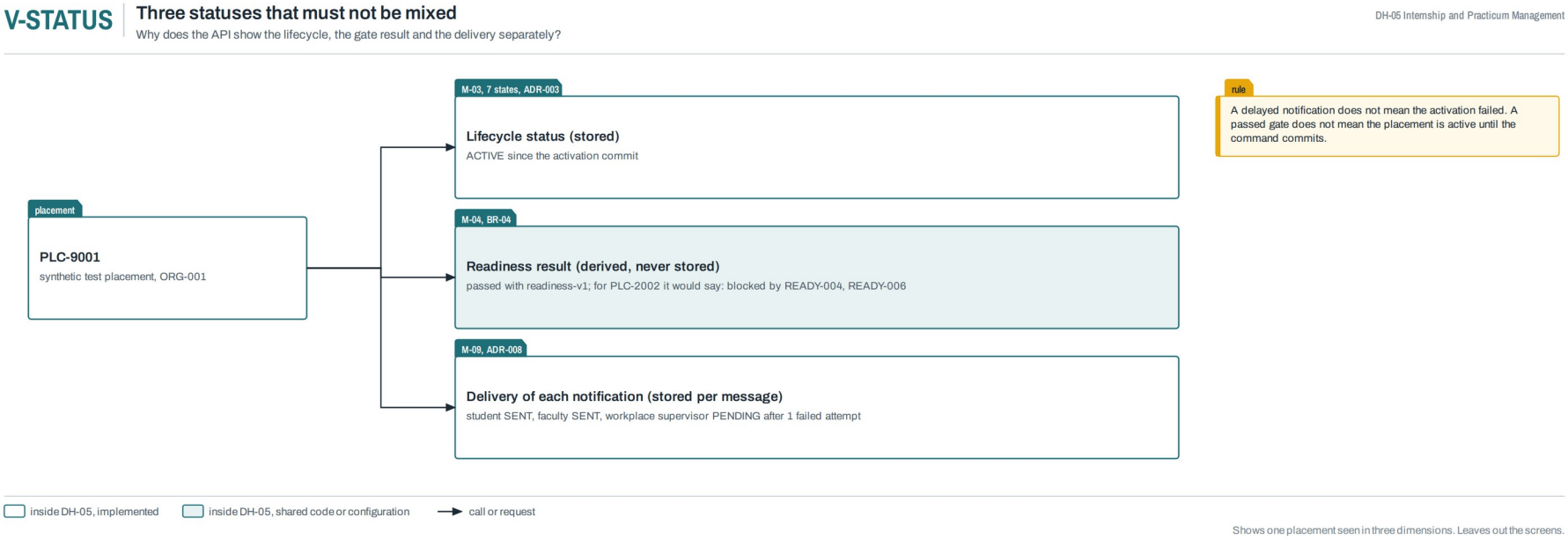


Figure 9. V-STATUS Lifecycle status, gate result and delivery status are separate facts.

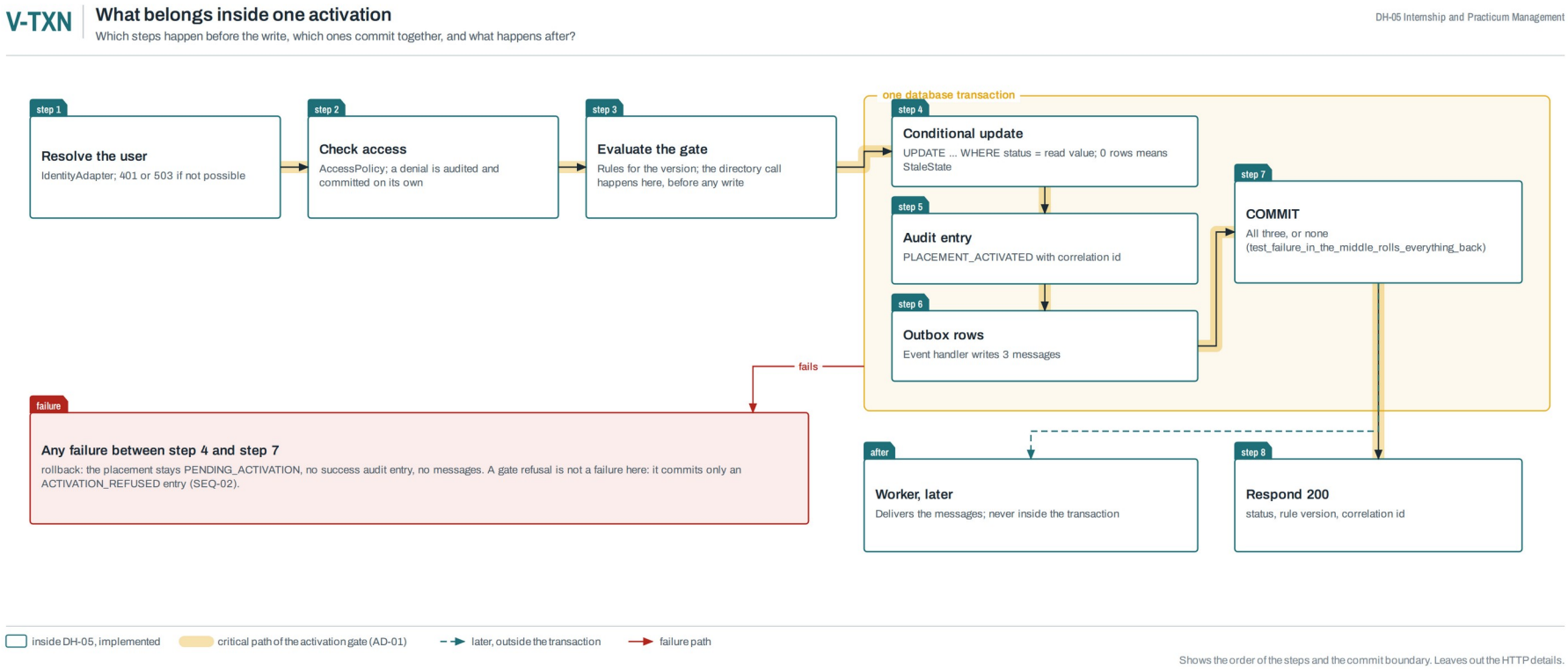


Figure 10. V-TXN What happens before the write, what commits together, and what happens after.

The directory call happens before the write phase, never while the write is pending. A refusal by the gate is not a failure of the transaction: it commits only an ACTIVATION_REFUSED entry.

Data lifecycle

Data	Rule
Placements	Never deleted; CANCELLED and CLOSED remain for program validation.
Audit entries	Append and read only through the application; a database administrator could still change rows (AR-05).
Outbox messages	SENT and FAILED are kept for diagnosis; an archive policy is not defined yet (BL-12).
Rule versions	Old versions stay in Git so every stored version label can be explained.
Personal data	Only synthetic data in the course (C-02). Real retention rules must be defined before a pilot.
Raw data pack	Never modified (INV-08); every transformation is written to data/generated/transformation_log.md.
Schema changes	create_all builds the schema today; versioned migrations are needed before shared data exists (BL-13).

8. Critical interactions and failure behavior

Structure is easy to draw; behavior under failure is where architectures break. Five interactions carry the drivers. Each one was drawn with its failure and each one has a test.

ID	Interaction	Drivers	Scenario	What it shows
SEQ-01	Activation succeeds	AD-01, AD-03, AD-05	QS-08	Access, gate, directory, transition, audit, event and outbox in one transaction.
SEQ-02	Activation refused by the gate	AD-01, AD-03	QS-08	Nothing changes except an ACTIVATION_REFUSED audit entry.
SEQ-03	Employer Directory is down	AD-05, AD-01	QS-08	Gate stays closed (fail closed).
SEQ-04	Two coordinators activate at the same time	AD-01, AD-03	QS-09	Conditional update: one wins, the other gets STATE_CHANGED.
SEQ-05	Notification delivery and failure	AD-05	QS-06	Worker delivers after commit; failures are retried, never undo the decision.

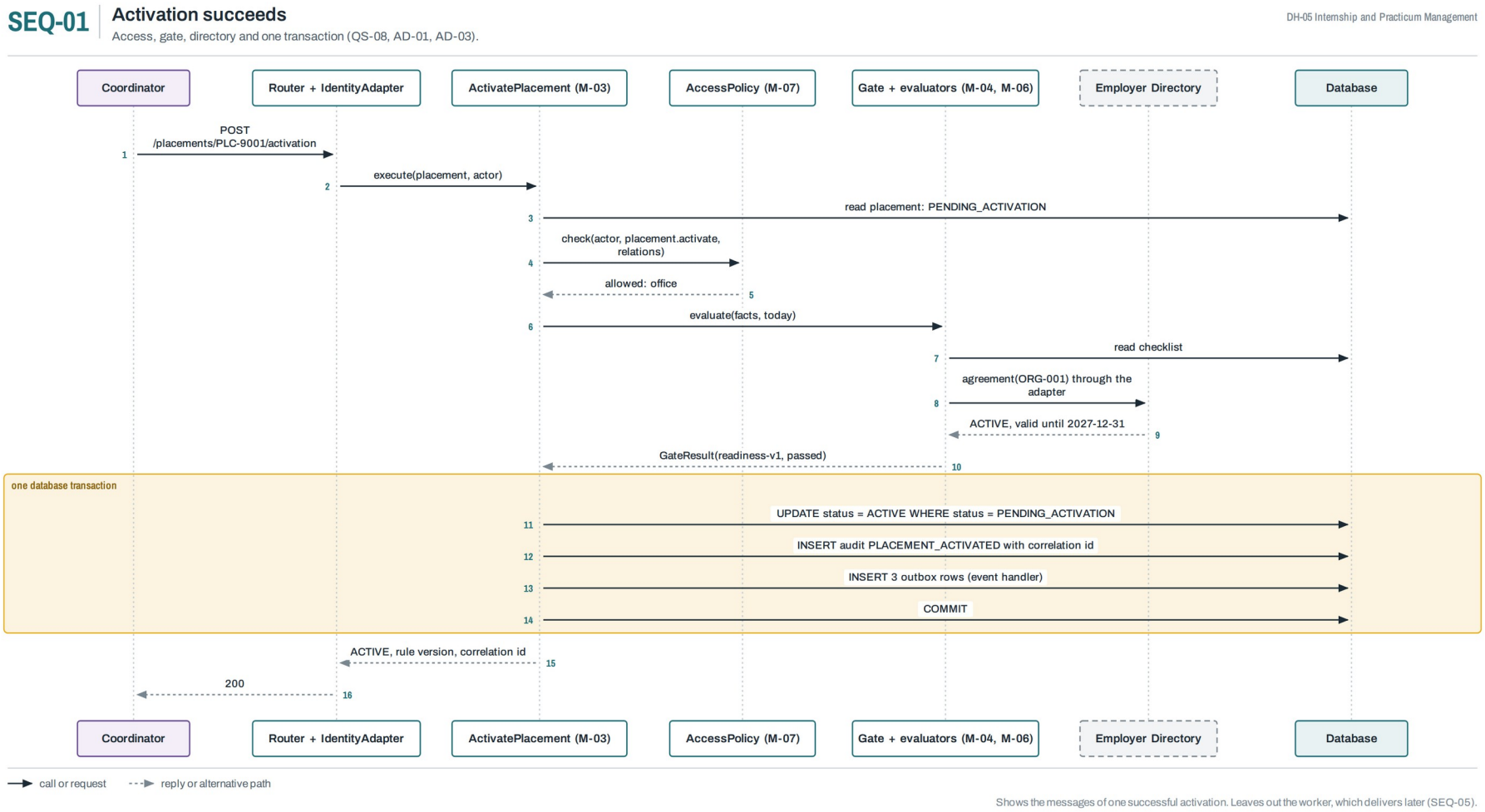


Figure 11. SEQ-01 Activation succeeds.

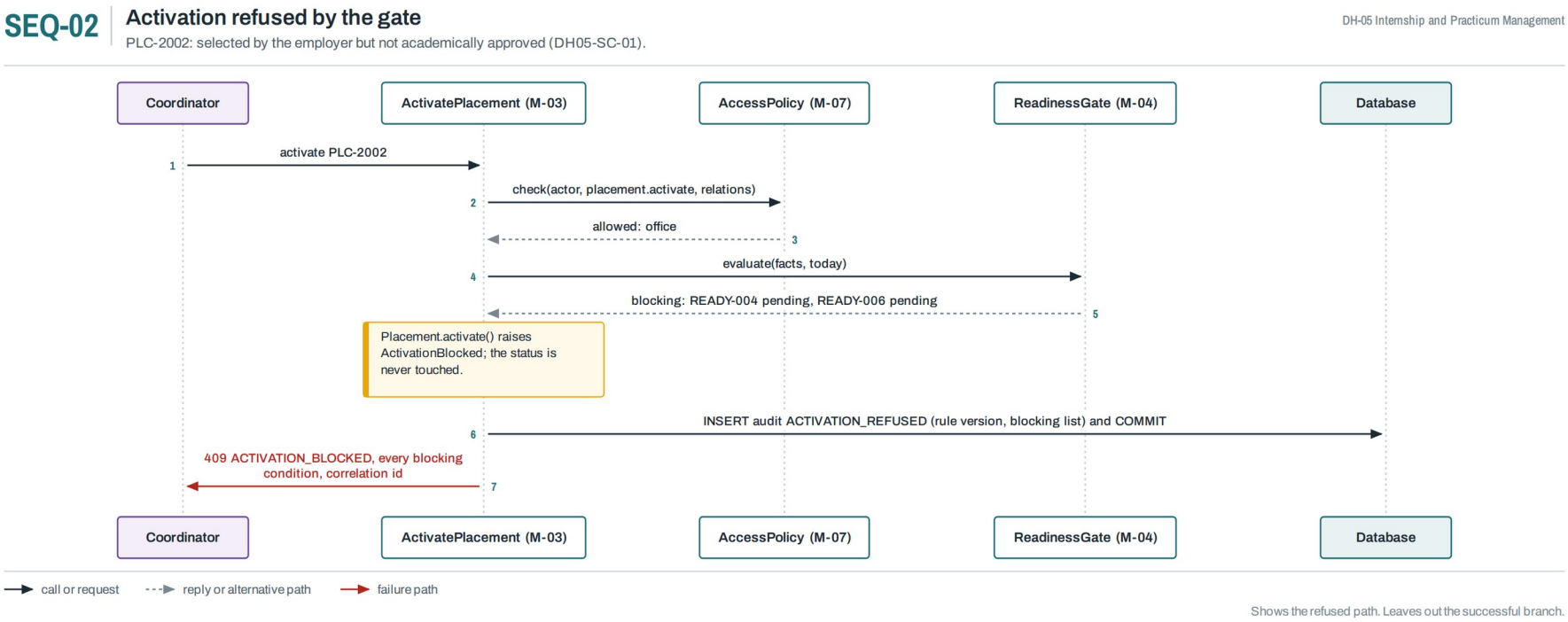


Figure 12. SEQ-02 Activation refused by the gate.

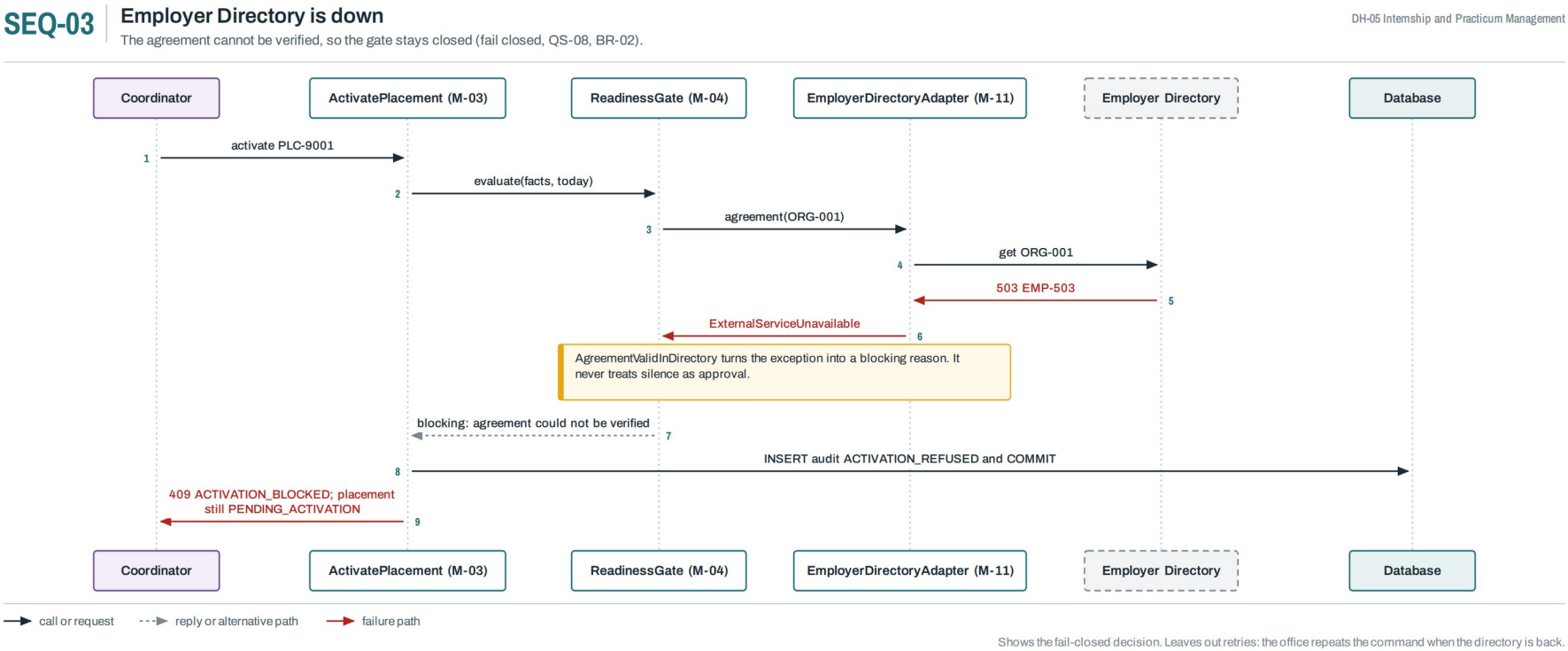


Figure 13. SEQ-03 Employer Directory is down: fail closed.

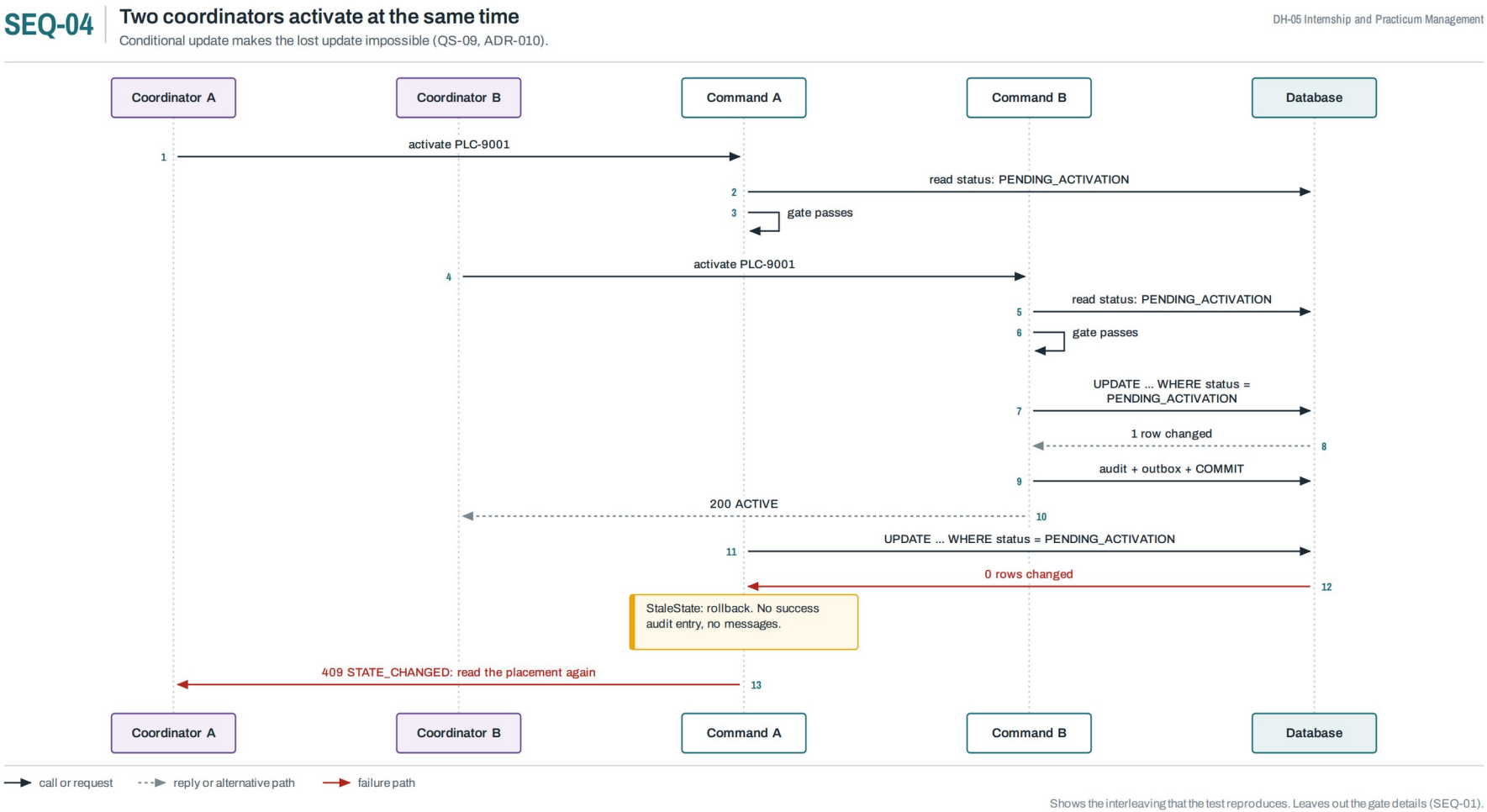


Figure 14. SEQ-04 Two coordinators activate at the same time (ADR-010).

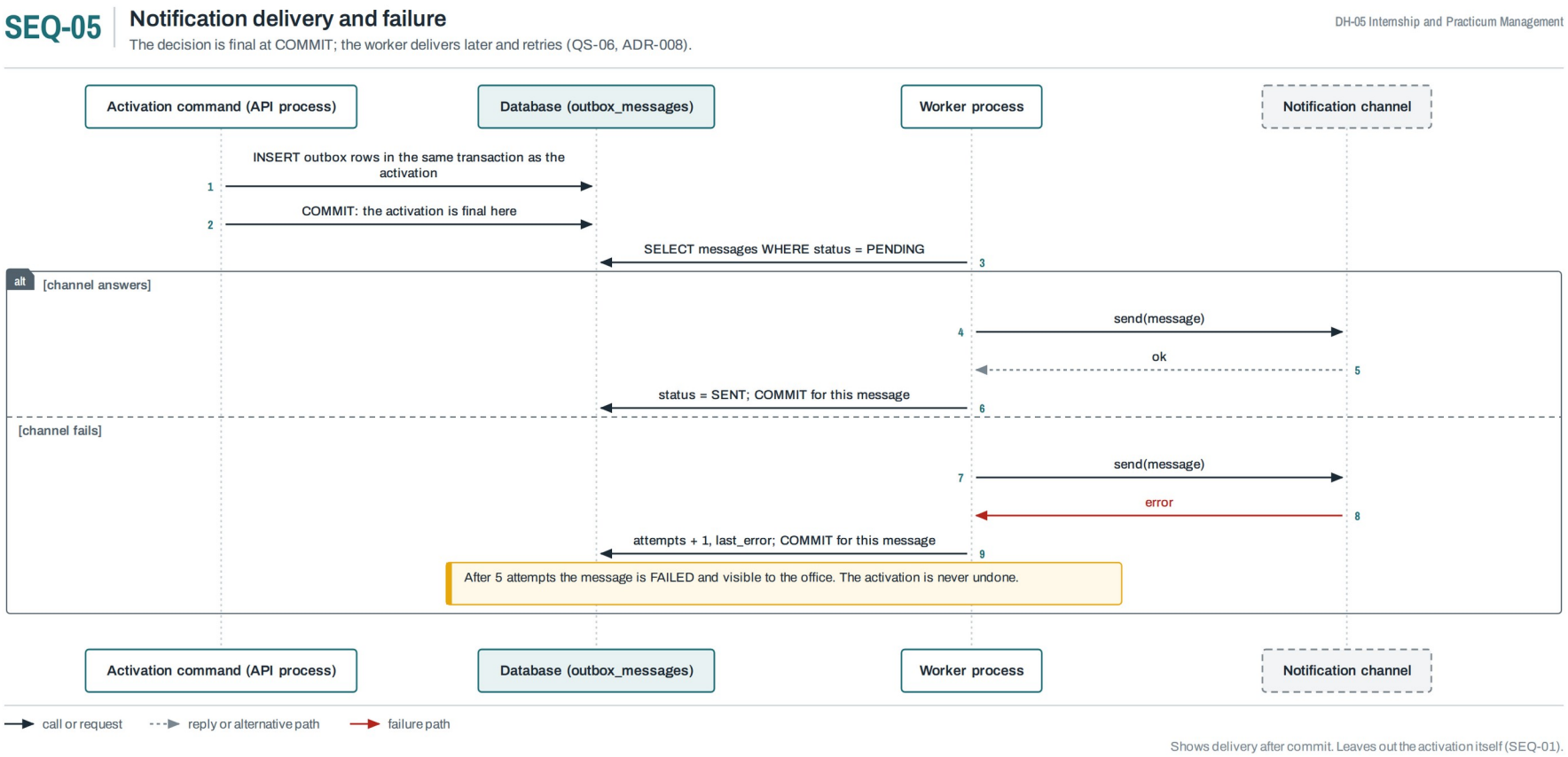


Figure 15. SEQ-05 Notification delivery and failure.

The failure matrix makes every claim reviewable: what fails, where it is detected, what the system does, what state is left, who sees what, and which test proves it. Rows marked NOT EXECUTED are honest gaps tracked in the backlog.

ID	What fails	Detected in	Behavior	State afterwards	Who sees what	Verified by
FM-01	Required condition PENDING, missing or expired	M-04 evaluator	Refuse: 409 ACTIVATION_ BLOCKED with every blocking condition	Placement unchanged; ACTIVATION_ REFUSED audit entry	Office sees the list and the correlation id	test_sc01_*, test_ expired_condition_ blocks_even_if_ marked_complete
FM-02	Employer Directory 503 / timeout	Agreement ValidIn Directory	Fail closed: refuse, reason 'could not be verified'	Placement unchanged	Office sees the reason and can retry	test_directory_down_ means_activation_ refused
FM-03	Employer Directory 409 (data conflict)	Agreement ValidIn Directory	Refuse, reason 'directory data in conflict, manual review'	Placement unchanged	Office is told to review manually	NOT EXECUTED for the activation path (unit-tested in adapter only)
FM-04	Employer Directory 404 (unknown organization)	Agreement ValidIn Directory	Refuse, reason 'organization unknown to the directory'	Placement unchanged	Office	NOT EXECUTED for the activation path
FM-05	Student Information 503 / 409 / 404	Student Information Adapter	Translated to port exceptions; M-02 policy: pending + retry / manual review / business error	Application kept	Student sees pending and reason	Adapter translation VERIFIED (3 cases); M-02 policy DESIGNED
FM-06	User without relationship	AccessPolicy	404 as if the placement did not exist; ACCESS_DENIED audited when the placement exists (an unknown id leaves nothing to audit)	Nothing changes	Caller learns nothing	test_unrelated_user_ gets_not_found, test_ qs02_*
FM-07	User with relationship but not allowed to act	AccessPolicy	403; audit ACCESS_DENIED	Nothing changes	Caller knows the placement exists	test_student_cannot_ activate_and_denial_ is_audited, test_host_ organization_cannot_ activate_and_denial_ is_audited
FM-08	Another command changed the status first	SqlPlacement Repository. save	409 STATE_CHANGED; rollback; no success audit, no messages	Winner's state stands	Loser is told to read again	test_two_concurrent_ activations_only_one_ commits (repository), test_api_answers_ state_changed_when_ another_command_ wins (409 through the API)
FM-09	Failure after the state change, before commit	Activate Placement. execute	Rollback of state, audit and outbox	Placement still PENDING_ ACTIVATION	Caller gets an error; retry is safe	test_failure_in_the_ middle_rolls_ everything_back
FM-10	Command repeated after success	Placement. activate	409 ILLEGAL_TRANSITION; no second audit entry or messages	Unchanged	Caller learns it is already active	test_repeating_the_ command_adds_no_ second_set_of_side_

ID	What fails	Detected in	Behavior	State afterwards	Who sees what	Verified by
						effects
FM-11	Notification channel fails	deliver_ pending	Retry up to 5 attempts, then FAILED	Activation committed; message PENDING/FAILED	Failure visible in the outbox	test_failed_delivery_ keeps_activation_and_ retries, test_outbox_ message_is_failed_ after_five_attempts
FM-12	Worker dies in the middle of a run	deliver_ pending	Commit per message: sent messages stay SENT	Remaining messages PENDING	Next run continues	test_qs06_worker_ crash_does_not_ resend_what_was_ already_sent
FM-13	Identity provider disabled or user unknown	get_current_ actor	503 if disabled; 401 if missing, unknown or inactive	Nothing changes	Caller	test_mock_identity_ disabled_answers_503, test_placements_ endpoint_requires_a_ known_identity
FM-14	Rule configuration file missing or invalid	Rule Configuration	Request fails with a server error; nothing is activated	Unchanged	Operator	NOT EXECUTED (no negative test yet; BL-06)

9. Interfaces and user-facing behavior

Modules talk in two ways: synchronous calls through a facade when the caller needs the answer, and domain events when the publisher does not need to know who reacts.

Domain event	Published by	Handled by	Reaction
ApplicationSubmitted	M-02	M-09	Notify the student and the employer contact
EligibilityDecided	M-02	M-09	Notify the student with the result and the rule version
EmployerDecisionRecorded	M-02	M-03, M-09	Create the placement in PENDING_ACTIVATION when the student is selected
AcademicApprovalRecorded	M-02	M-04	Complete the ACADEMIC_APPROVAL readiness condition
WorkPlanApproved	M-03	M-04, M-09	Complete the WORK_PLAN readiness condition
ReadinessConditionChanged	M-04	M-09, M-10	Remind the responsible role; update at-risk report
PlacementActivated	M-03	M-09	Notify student and both supervisors
PlacementStarted	M-03	M-09	Start hour and evaluation reminders
HoursReviewed	M-05	M-09	Notify the student of a confirmation or dispute
EvaluationOverdue	M-09 (scheduler)	M-09, M-10	Reminder to the evaluator; at-risk report
PlacementClosed	M-03	M-09, M-10	Notify participants; update reports

Endpoint catalog. Only the two activation-slice operations are implemented; the rest is the proposed contract for Evaluation 3.

API operation	Module	Allowed caller	Stories
POST /opportunities	M-01	Host organization contact	US-002
POST /opportunities/{id}/review	M-01	Experiential-learning coordinator	US-003
GET /opportunities?program=	M-01	Student	US-004
POST /applications	M-02	Student	US-005, US-007
POST /applications/{id}/employer-decision	M-02	Host organization contact	US-009
POST /applications/{id}/academic-approval	M-02	Academic program coordinator	US-011
POST /placements/{id}/work-plans	M-03	Academic program coordinator	US-010, US-021
GET /placements/{id}/readiness	M-04	Related users only: office, faculty supervisor, workplace supervisor, student, host organization (projection)	US-012
POST /placements/{id}/activation	M-03	Experiential-learning coordinator	US-014, US-016
POST /placements/{id}/hours	M-05	Student	US-017
POST /hours/{id}/review	M-05	Workplace supervisor	US-018
POST /placements/{id}/closure	M-03	Experiential-learning coordinator	US-025
GET /placements/{id}/history	M-08	Office, auditor	US-026

Contract: POST /placements/{placement_id}/activation (IMPLEMENTED)

Stories	US-014, US-016
Allowed caller	Experiential-learning coordinator (staff role EXPERIENTIAL_LEARNING_COORDINATOR)
Request	Header X-Mock-User-Id (required in development). No body.
Idempotency	Not idempotent by design: a second call after success returns 409 ILLEGAL_TRANSITION and adds no side effects (FM-10).
Versioning	No version prefix yet; breaking changes require a new ADR (BL-08).

Status	Body	Meaning
200	{placement_id, status: 'ACTIVE', rule_version, correlation_ id}	Gate passed; state, audit and outbox committed.

Status	Body	Meaning
401	{detail}	Header missing, user unknown or inactive.
403	{detail}	User related to the placement but not allowed to activate. Audited.
404	{detail}	Placement does not exist OR user has no relationship (indistinguishable). Audited only when the placement exists.
409	{code: ACTIVATION_BLOCKED, detail, blocking: [{requirement, check_id, reason}], correlation_id}	At least one rule blocks. Audited as ACTIVATION_ REFUSED.
409	{code: STATE_CHANGED, detail}	Another command changed the placement first. Read it again.
409	{code: ILLEGAL_TRANSITION, detail}	Placement is not in PENDING_ACTIVATION.
503	{detail}	Identity provider disabled.

Contract: GET /placements/{placement_id}/readiness (IMPLEMENTED)

Stories	US-012
Allowed caller	Office, faculty supervisor, workplace supervisor, student, host organization (related users only)
Request	Header X-Mock-User-Id.
Idempotency	Safe read; the gate is evaluated at read time, so the result can change between calls.
Versioning	As above.

Status	Body	Meaning
200	{placement_id, status, audience, rule_version, ready, blocking[], conditions[{check_id, requirement_type, status, evidence_reference}]}	evidence_reference is null for RESTRICTED requirements outside the office.
401	{detail}	As above.
404	{detail}	Not found or no relationship. Denial audited when the placement exists.
503	{detail}	Identity provider disabled.

Errors tell the caller how to recover

Status	Code	Meaning	Recovery
401	-	Header missing, user unknown or inactive	Sign in again; do not retry automatically.
403	-	Related to the placement but not allowed to act	Do not retry; ask the office.
404	-	Placement missing or no relationship (indistinguishable on purpose)	Check the identifier; no retry.
409	ACTIVATION_BLOCKED	At least one rule blocks; the list is in the body	Complete the named conditions, then activate again.
409	STATE_CHANGED	Another command changed the placement first	Read the placement again; do not resend blindly.
409	ILLEGAL_TRANSITION	The placement is not in PENDING_ ACTIVATION	Read the current status; nothing to do if already ACTIVE.
503	-	Identity provider disabled	Retry after the identity service is configured.

Event, outbox and port contracts

The event as it is published today (in process):

```
{
  "name": "PlacementActivated",
  "occurred_at": "2026-09-24T14:05:00Z",
  "payload": {
    "placement_id": "PLC-9001",
    "student_id": "HU-STU-5002",
    "workplace_supervisor_id": "SUP-EXT-002",
    "faculty_supervisor_id": "FAC-SUP-101",
    "correlation_id": "<uuid hex, one per command>"
  }
}
```

The current event has no schema version and no event id. It never leaves the process, so this is acceptable today; before a module is extracted or an external consumer appears, add eventId and schemaVersion (BL-11).

Contract	Owner	Content	Guarantees and limits
Placement Activated	M-03	payload: placement_id, student_id, workplace_supervisor_id, faculty_ supervisor_id, correlation_id	Handlers run inside the command's transaction and may only write local rows. Order of handlers is not guaranteed. Delivery is in-process and at-most-once: a crash before commit means the event never happened.

Contract	Owner	Content	Guarantees and limits
Outbox message	M-09	event_name, recipient_id, payload {placement_id, template}, status PENDING/ SENT/FAILED, attempts, correlation_id	At-least-once towards the channel: a crash between send and commit may repeat one message; the receiver must tolerate a duplicate (documented limitation).

Port	Failure contract
EmployerDirectoryPort.agreement(organization_id) -> AgreementSummary	Raises ExternalServiceUnavailable (503), ExternalDataConflict (409), ExternalRecordNotFound (404). Implemented by Employer DirectoryAdapter (M-11).
StudentInformationPort.profile(student_id) -> Student Profile	Same three exceptions. Carries read_at. Implemented by Student InformationAdapter (M-11).
IdentityPort.resolve(user_id) -> Identity | None	None for unknown users. Implemented by IdentityAdapter reading mock_identities.json.

Integrations

For each external system we answered the eight questions of the course integration checklist (full answers in the Integration Design sheet). Failure behavior is a business decision: fail soft when a request can wait, fail closed when going ahead without the answer would put the student and the university at risk (for example, activating without a valid agreement).

System	Authoritative for	Communication	Failure behavior
Identity service (simulated)	Identity service owns user id, display name, e-mail, institutional role and active flag.	Header X-Mock-User-Id resolved by the Identity Adapter (reads mock_ identities.json). Real SSO later.	Fail closed: no identity, no access.
Student Information Service (simulated)	Authoritative for program, period, enrollment, credits, average, standing and previous placements (ASM-002).	In-process call to the mock service class through Student InformationAdapter (same payload as the contract GET / mock/student-information/ students/{student_id}); HTTP when the real service exists.	404 -> business error; 409 -> manual review; 503 -> pending + retry (fail soft).
Employer Directory Service (simulated)	Authoritative for organization identity, active status, sector, country and agreement summary (ASM-003).	In-process call to the mock service class through EmployerDirectoryAdapter (same payload as the contract GET / mock/employers/{organizatio n_id}); HTTP when the real	404 -> activation refused, 'organization unknown to the directory'; 409 -> activation refused, 'directory data in conflict, manual review' (FM-03); 503 -> activation refused,

System	Authoritative for	Communication	Failure behavior
		service exists.	'could not be verified' (FM-02).
Notification channel (simulated)	DH-05 owns the message content; the channel only delivers.	Outbox table read by the worker process.	Retry up to 5 attempts, then FAILED and visible in the outbox; an alert to the office is DESIGNED.

User-facing behavior (designed)

Interaction	Design rule
Readiness panel	Show every blocking condition at once with its id and the role that must act; never only the first one.
Activate button	Disabled with the reason while the gate blocks; the API still decides (BR-01).
Conflict	On 409 STATE_CHANGED, reload the placement and explain that someone else acted; keep no stale data on screen.
Restricted evidence	External audiences see 'Complete (reference hidden)', never an empty field that looks like a bug.
Status words	Lifecycle, gate result and delivery are three labels, written in words, not color alone.
Double click	The UI may prevent it, but correctness comes from conditional update and ILLEGAL_ TRANSITION.

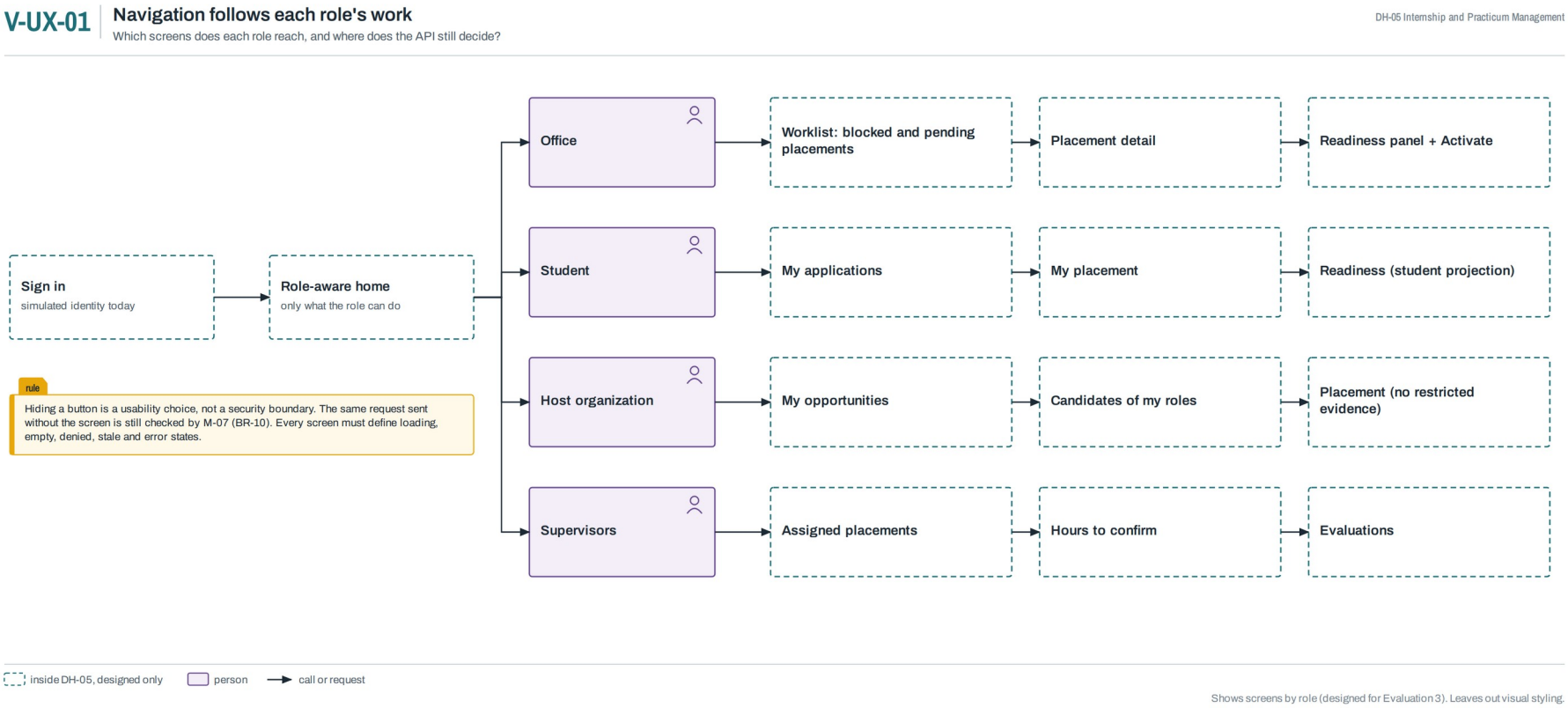


Figure 16. V-UX-01 Navigation by role.

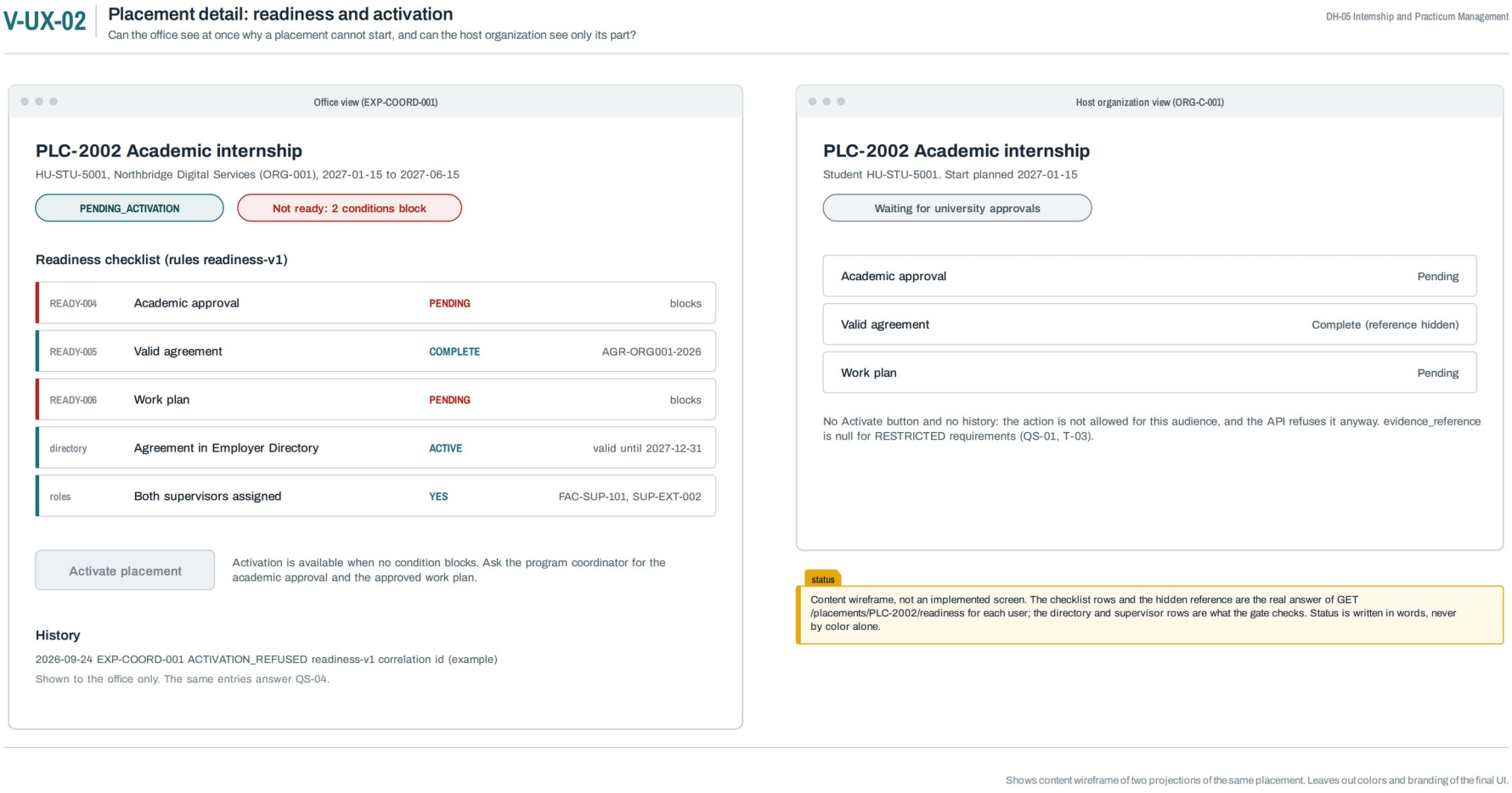


Figure 17. V-UX-02 Content wireframe of the placement detail for two audiences.

The office sees every blocking condition and why Activate is disabled; the host organization sees only its projection. The usability check for this screen is planned, not run.

10. Security and privacy

Access is decided in one place (M-07) with role plus relationship, and every response goes through an audience projection. Authentication only establishes who the caller is; it never authorizes an action by itself. The matrix is deliberately conservative: where CL-018 is open, access is denied until the Data Privacy Officer answers.

Resource	Student	Host org.	Workplace sup.	Faculty sup.	Program coord. (DESIGNED)	Office
Published opportunity	Read if their program is eligible	Read/edit own organization's	-	Read	Read	Read/review
Application (status, documents APPLICATION)	Own applications	Candidates of own opportunities	-	-	Own program	All
Student academic profile (credits, average, standing)	Own	Never	Never	Own placements	Own program	All
Placement summary and readiness checklist	Own	Own organization, without RESTRICTED evidence	Assigned placements	Assigned placements	Own program	All
RESTRICTED documents (agreement, insurance, confidentiality)	Status only	Status only	-	Status only	Status only	Full
Hour entries	Own (all levels)	-	Assigned placements	Assigned placements	Own program	All
Evaluation - student-visible feedback and rating	Own	Own evaluations submitted	Own evaluations	Assigned placements	Own program	All
Evaluation - restricted comments	Never	Never (CL-018 open)	Own comments only	Assigned placements	Pending CL-018 (deny)	All
Incident with visibility STUDENT/PLACEMENT_TEAM	If reporter or placement member	-	Assigned placements	Assigned placements	Own program	All
Incident with visibility RESTRICTED or INTERNAL	Only if they reported it	Never	Never (CL-018 open)	Assigned placements	Pending CL-018 (deny)	All
Audit history	-	-	-	-	Own program (read)	All (read)

Implemented today (activation slice): 'office' means the staff roles EXPERIENTIAL_LEARNING_COORDINATOR, PROGRAM_DIRECTOR and AGREEMENTS_REPRESENTATIVE, and only the first may activate. The academic program

coordinator column is DESIGNED: the relationship 'coordinator of the program' is not implemented yet. The other rows describe the target policy for the modules of Evaluation 3.

Asset	Why it matters
Student academic profile and readiness evidence references	Privacy of students (QR-01)
Restricted evaluation comments and sensitive incidents	Dignity and safety (QR-02)
Placement status and audit history	Integrity of the process (QR-03, AD-01)
Rule configuration	Integrity of the gates (AD-04)

ID	Boundary	What crosses and how we treat it
TB-1	Browser / client -> API	Untrusted input crosses here. Identity is a header in development (DA-04); real authentication is not designed yet.
TB-2	API -> institutional services (identity, student, employer)	Their answers are data, not truth about our rules; they may fail or be stale; adapters translate and fail closed where needed.
TB-3	API -> database and rule files	Trusted zone; access only with the application credentials; rule files are reviewed in version control.
TB-4	Worker -> notification channel	Messages leave the system; they must contain no restricted data.

A small threat model is better than 'we use HTTPS'. We list the threats we do not mitigate as well as the ones we do; the identity header (T-01) is acceptable for a course project and not for a pilot. This is a scoped review, not a penetration test.

ID	Category	Threat	Control	Status	Evidence
T-01	Spoofing	A caller sends any X-Mock-User-Id and acts as a coordinator.	Development-only mechanism (DA-04). Production needs real authentication; ports isolate the change. Risk accepted for the course and NOT acceptable for a pilot.	DESIGNED (open risk AR-02)	ADR-007; technical-assumptions.md
T-02	Information disclosure	Enumerating placement ids reveals which exist.	Unrelated users get 404, identical to a missing placement.	VERIFIED	test_unrelated_user_gets_not_found
T-03	Information disclosure	Host organization or student reads RESTRICTED evidence references (agreement, insurance).	Audience projection removes evidence_reference outside the office.	VERIFIED	test_host_organization_sees_readiness_without_restricted_evidence
T-04	Information disclosure	A notification carries restricted data to an external recipient.	Outbox payload contains only placement id and template.	VERIFIED	test_qs06_outbox_payload_has_no_restricted_

ID	Category	Threat	Control	Status	Evidence
					data
T-05	Tampering	Someone edits or deletes audit history.	No update/delete interface; application never exposes one. A database administrator could still alter rows: not prevented.	VERIFIED (interface) / open (DB-level)	test_audit_history_can_only_be_appended_and_read; AR-05
T-06	Tampering	A weaker rule version is published to let a placement start.	Rules are files in version control reviewed by the team; version is stored in each decision. No runtime admin screen yet.	DESIGNED	ADR-004; BL-07
T-07	Repudiation	A coordinator denies having activated a placement.	Audit entry with actor, time, previous and new value, rule version and correlation id.	VERIFIED	test_ready_placement_is_activated_audited_and_notified
T-08	Elevation of privilege	A student or external user calls the activation endpoint.	Access policy allows only the coordinator staff role; denial is audited.	VERIFIED	test_student_cannot_activate_and_denial_is_audited
T-09	Tampering / integrity	Two concurrent commands produce two success records or two sets of notifications.	Conditional update on status; single transaction.	VERIFIED (deterministic interleaving on SQLite)	test_two_concurrent_activations_only_one_commits
T-10	Denial of service	A flood of requests exhausts the application or the directory.	Not addressed: load is small (DA-01) and no rate limiting is designed.	NOT DESIGNED (accepted, AR-06)	-

11. Deployment and operations

Two environments matter: the demonstration environment that runs today on one computer, and the pilot environment that is only designed. Figure 18 at the end of this section shows both, with the trust boundaries.

Element	Choice	Why	Decision
Language and framework	Python 3.12, FastAPI	Supplied by the starter, known by the team, small and maintainable (C-03, C-06).	ADR-000
Persistence	SQLAlchemy; SQLite locally, PostgreSQL prepared	One relational database with local transactions (AD-01, AD-03).	ADR-002
Rules	Versioned JSON files + Strategy evaluators	Rules change by program without redeploy (AD-04).	ADR-004
Integration	Ports and adapters over the DH-05 mocks	Services are authoritative and fail (AD-05).	ADR-007

Element	Choice	Why	Decision
Side effects	In-process events + outbox table + worker	Notifications must never undo a decision (QR-04).	ADR-008
Concurrency	Conditional update on status, correlation id	One winner per placement (QS-09).	ADR-010
CI	GitHub Actions: ruff, data validation, pytest	Configured to integrate often and keep main green; it has not run yet (no remote).	Section 12
Web client	Not decided	No driver depends on it; decided in Evaluation 3.	-

Environmental assumptions

Concern	Assumption or configuration
Runtime	Python 3.12, FastAPI, SQLAlchemy 2; versions recorded in docs/evaluation-2/validation/versions.txt.
Database	SQLite file by default; PostgreSQL 16 profile in docker-compose.yml, not yet exercised by the suite (TP-14).
Schema	Created by create_all at start-up; no migrations yet (BL-13).
Mocks	In-process classes reading mocks/*.json; failure injected with MOCK_EMPLOYER_FAILURE=EMP-503.
Identity	Header X-Mock-User-Id with MOCK_IDENTITY_ENABLED=true; development only (DA-04).
Worker	Run on demand with scripts/run_worker.py; a scheduler is a deployment decision for the pilot.
Data	DH-05 release-1 data pack; manifest hashes verified (placement_types.csv unchanged).
Reproducibility	pip install -e '.[dev]'; python scripts/seed_domain.py; pytest. The same commands are configured in CI (.github/workflows/ci.yml), which has not run yet.

Signals to diagnose the main failures

Signal	Interpretation	Source	Status
Activation outcomes by code	Separates normal blocking from failures	Audit entries by action (ACTIVATION_ REFUSED, PLACEMENT_ACTIVATED, ACCESS_DENIED); query by resource_id	Available through the audit table
Oldest PENDING outbox message	Delivery is not progressing	outbox_messages.created_at where status = PENDING	Available (query); no alert yet
FAILED messages	Human action needed	outbox_messages.status = FAILED with last_ error	Available (query)
STATE_CHANGED rate	Real contention between	API responses with code STATE_CHANGED	DESIGNED (no

Signal	Interpretation	Source	Status
	coordinators		metrics endpoint)
Directory failures	Activations blocked by an outage	Blocking reason 'could not be verified' in refusals	Available in audit new_value
Correlation	Follow one command end to end	correlation_id in audit, outbox and response	Available

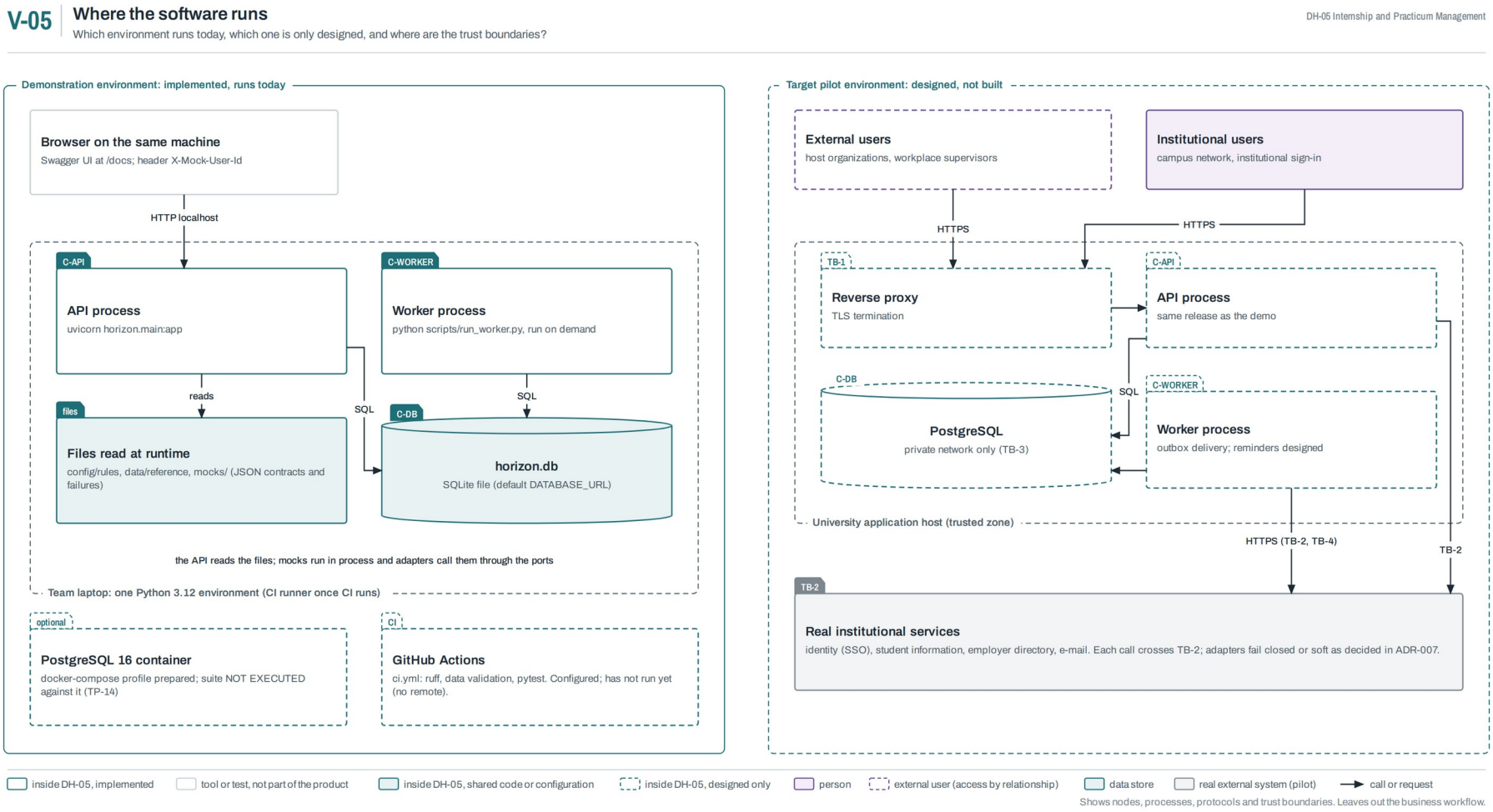


Figure 18. V-05 Deployment view: the demonstration environment that runs today and the pilot environment that is only designed.

No container image or shared server exists yet; we do not claim one. The pilot view states what would have to be true before real users touch the system.

12. Validation: test strategy, evidence and status

Evaluation 2 does not ask for the working system; that is Evaluation 3. It asks whether the structure is feasible. Our proof of concept attacks the riskiest assumption of the design: that the activation gate cannot be bypassed, even under concurrency, partial failure or a failing directory. It uses the real relational mechanism, not an in-memory substitute.

ID	Level	Intent	Covers	How	Status
TP-01	Unit	Lifecycle transition table and guards	BR-01, BR-03, INV-01..03	tests/unit/test_placement_lifecycle.py (6 tests)	VERIFIED
TP-02	Unit	Rule evaluators and configuration-only rule change	BR-01, BR-08, QS-07	tests/unit/test_readiness_rules.py (5 tests)	VERIFIED
TP-03	Unit	Adapters translate payloads and failures; program aliases	AD-05, QS-05	tests/unit/test_reference_and_adapters.py (7 cases)	VERIFIED
TP-04	Integration	Activation slice end to end through the API	QS-08, BR-01, BR-02	tests/integration/test_activation_endpoint.py (8 tests)	VERIFIED
TP-05	Integration	Concurrency, atomicity, repetition, correlation, machine-readable codes	QS-09, QS-04, BR-12, INV-04, INV-05, INV-07	tests/integration/test_concurrency_and_atomicity.py (5 tests)	VERIFIED
TP-06	Integration	Outbox: failed channel, retry, crash, payload content	QS-06, BR-11	test_outbox.py (1) + test_traceability_evidence.py (3: payload, crash, FAILED after 5)	VERIFIED
TP-07	Integration	Access: unrelated user, wrong role, supervisor relationship, projection	QS-02, QS-01 (partial), BR-09, BR-10	activation tests + test_traceability_evidence.py (401, 403 host organization, student projection; workplace supervisor at policy level, EU-05)	VERIFIED (slice only)
TP-08	Architecture	Dependency rules, purity of domain, mocks only through M-11, audit append-only	ADR-001, ADR-007, ADR-006	tests/architecture/test_module_boundaries.py (5 tests)	VERIFIED
TP-09	Data	Seed keeps raw files intact and loads every placement	EU-03, INV-08	test_traceability_evidence.py (2 tests); scripts/validate_data.py (run locally)	VERIFIED
TP-10	Mutation check	Do the tests detect real defects? Six defects were injected one at a time into a copy of the code (no conditional update, directory failure lets the gate pass, projection leaks evidence, student may activate, forbidden import, worker commits once per run) and each made at least one test fail	Test quality	docs/evaluation-2/validation/mutation_check.py and mutation_output.txt	EXECUTED (6 of 6 detected); script is repeatable, not part of CI
TP-11	Contract	Mock service contracts against the adapters	ADR-007	Adapter tests use the real mock classes and failure ids; no consumer-driven contract tool	PARTIAL
TP-12	Acceptance	Eligibility, employer decision, academic approval, hours,	QS-01, QS-03, QS-05,	Planned with the modules of	NOT EXECUTED - modules not

ID	Level	Intent	Covers	How	Status
		evaluation, closure scenarios	BR-05, BR-06	Evaluation 3	built
TP-13	Non-functional	Load at the seasonal peak (QS performance is a non-driver)	DA-01	Not planned unless DA-01 is contradicted	NOT EXECUTED - by decision
TP-14	Environment	Same suite against PostgreSQL; true parallel concurrency	DA-03, QS-09	Run docker-compose profile with two real connections	NOT EXECUTED - no PostgreSQL in the authoring environment
TP-15	Security	Authentication, rate limiting, dependency scan	T-01, T-10	Needs real authentication first	NOT EXECUTED - not designed

Validation evidence records

Executed on 2026-10-11 with the versions in docs/evaluation-2/validation/versions.txt: 59 tests passed, 0 failed; ruff check reported no issues. The records below separate the expected result from what we observed.

Test	TP-01 / tests/unit/test_placement_lifecycle.py (6 tests)
Claim	BR-01, BR-03, INV-01..03: the lifecycle only moves through the transition table; ACTIVE only through activate() with a passed gate.
Status	PASSED
Setup	Placement objects built in memory; no database.
Expected	Activation without a passed gate is refused; other commands cannot reach ACTIVE; closing from PENDING_ACTIVATION is illegal; start date only after activation; interrupted closure allowed from ON_HOLD.
Observed	As expected (6 passed).
Limitations	Domain level only; persistence and API are covered by TP-04 and TP-05.
Related	requirements: US-014; qualities: QS-08; decisions: ADR-003
Fixture	None: Placement objects built in memory
Command	python -m pytest -o addopts="" -v (from the repository root)
Environment	Windows 11, Python 3.12.1; no database
Independent connections	not applicable
Executor	Sebastián Mercado, 2026-10-11
Executed at	2026-10-11
Trials	1 run; the tests are deterministic

Raw output	docs/evaluation-2/validation/pytest_output.txt
Source digest	f7281a5c6c9e7a50 (command: find src tests config -type f \(-name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' | LC_ALL=C sort | xargs sha256sum | sha256sum | cut -c1-16)

Test	TP-02 / tests/unit/test_readiness_rules.py (5 tests)
Claim	BR-01, BR-02, BR-08, QS-07: pending, expired or missing conditions block; directory failure keeps the gate closed; a rule change is configuration only.
Status	PASSED
Setup	Rule evaluators with in-memory conditions; a temporary copy of readiness.json for the configuration test.
Expected	Each blocking case returns a blocking result; adding CONFIDENTIALITY_FORM for one program changes the result without code changes.
Observed	As expected (5 passed).
Limitations	Readiness rules only; eligibility rule types are designed.
Related	requirements: US-012, US-014; qualities: QS-07, QS-08; decisions: ADR-004
Fixture	In-memory conditions; temporary copy of config/rules/readiness.json
Command	python -m pytest -o addopts="" -v (from the repository root)
Environment	Windows 11, Python 3.12.1; no database
Independent connections	not applicable
Executor	Sebastián Mercado, 2026-10-11
Executed at	2026-10-11
Trials	1 run; the tests are deterministic
Raw output	docs/evaluation-2/validation/pytest_output.txt
Source digest	f7281a5c6c9e7a50 (command: find src tests config -type f \(-name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' | LC_ALL=C sort | xargs sha256sum | sha256sum | cut -c1-16)

Test	TP-03 / tests/unit/test_reference_and_adapters.py (7 cases)
Claim	AD-05, QS-05, EU-03: adapters translate payloads and the three failures; the legacy program code is normalized.
Status	PASSED
Setup	The real mock service classes with their documented failure ids (SIS-404, SIS-409, SIS-503).

Expected	Failures become port exceptions; read_at is recorded; SWE becomes BSC-SWE; expired agreements are not valid.
Observed	As expected (7 passed).
Limitations	Mocks, not the real institutional services; no consumer-driven contract tool.
Related	requirements: US-012; qualities: QS-05; decisions: ADR-007
Fixture	The mock service classes and their JSON contracts in mocks/
Command	python -m pytest -o addopts="" -v (from the repository root)
Environment	Windows 11, Python 3.12.1; no database
Independent connections	not applicable
Executor	Sebastián Mercado, 2026-10-11
Executed at	2026-10-11
Trials	1 run; the tests are deterministic
Raw output	docs/evaluation-2/validation/pytest_output.txt
Source digest	f7281a5c6c9e7a50 (command: find src tests config -type f \(-name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' | LC_ALL=C sort | xargs sha256sum | sha256sum | cut -c1-16)

Test	TP-04 / test_sc01_*, test_sc02_*, test_directory_down_*
Claim	QS-08, BR-01, BR-02: activation refused when a condition blocks or the directory is down.
Status	PASSED
Setup	Seeded data pack (PLC-2002, PLC-2003); synthetic PLC-9001 with EMP-503 injected.
Expected	409 with READY-004 and READY-006; 409 with exactly five blocking results for PLC-2003 (READY-007 expired, READY-008 missing, directory agreement EXPIRED valid until 2026-07-31, ACADEMIC_APPROVAL and WORK_PLAN not registered); 409 'could not be verified'; status unchanged.
Observed	As expected.
Limitations	Mocked directory; the real service contract may differ.
Related	requirements: US-014, US-016; qualities: QS-08; decisions: ADR-003, ADR-004, ADR-007
Fixture	DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated
Command	python -m pytest -o addopts="" -v (from the repository root)
Environment	Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema

	created with create_all (no migrations, BL-13); versions in versions.txt
Independent connections	not applicable
Executor	Sebastián Mercado, 2026-10-11
Executed at	2026-10-11
Trials	1 run; the tests are deterministic
Raw output	docs/evaluation-2/validation/pytest_output.txt
Source digest	f7281a5c6c9e7a50 (command: find src tests config -type f \(-name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' | LC_ALL=C sort | xargs sha256sum | sha256sum | cut -c1-16)

Test	TP-05 / test_failure_in_the_middle_rolls_everything_back
Claim	INV-05: state, audit and outbox commit together or not at all.
Status	PASSED
Setup	Event handler writes the outbox rows and then raises.
Expected	Status PENDING_ACTIVATION, no success audit entry, no outbox rows (checked from a new session).
Observed	As expected.
Limitations	Shows nothing was committed; does not distinguish an explicit rollback from a discarded session.
Related	requirements: US-014; qualities: QS-04, QS-09; decisions: ADR-002, ADR-010
Fixture	DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated
Command	python -m pytest -o addopts="" -v (from the repository root)
Environment	Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema created with create_all (no migrations, BL-13); versions in versions.txt
Independent connections	not applicable
Executor	Sebastián Mercado, 2026-10-11
Executed at	2026-10-11
Trials	1 run; the tests are deterministic
Raw output	docs/evaluation-2/validation/pytest_output.txt
Source digest	f7281a5c6c9e7a50 (command: find src tests config -type f \(-name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' | LC_ALL=C sort | xargs sha256sum | sha256sum | cut -c1-16)

Test	TP-05 / test_two_concurrent_activations_only_one_commits
Claim	QS-09, BR-12, ADR-010: two commands that read the same status cannot both commit.
Status	PASSED
Setup	PLC-9001 with every condition complete; two SQLAlchemy sessions on the same SQLite file; the second command runs to completion between the first command's read and write.
Expected	First writer commits ACTIVE; second raises StaleState; one PLACEMENT_ACTIVATED entry; three outbox rows.
Observed	As expected (pytest output, docs/evaluation-2/validation/pytest_output.txt).
Limitations	Deterministic interleaving on SQLite, one trial; not parallel connections on PostgreSQL (TP-14).
Related	requirements: US-014; qualities: QS-04, QS-09; decisions: ADR-002, ADR-010
Fixture	DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated
Command	python -m pytest -o addopts="" -v (from the repository root)
Environment	Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema created with create_all (no migrations, BL-13); versions in versions.txt
Independent connections	2 SQLAlchemy sessions on the same SQLite file, deterministic interleaving
Executor	Sebastián Mercado, 2026-10-11
Executed at	2026-10-11
Trials	1 run; the tests are deterministic
Raw output	docs/evaluation-2/validation/pytest_output.txt
Source digest	f7281a5c6c9e7a50 (command: find src tests config -type f \(-name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' | LC_ALL=C sort | xargs sha256sum | sha256sum | cut -c1-16)

Test	TP-06 / test_failed_delivery_keeps_activation_and_retries, test_qs06_*, test_outbox_message_is_failed_after_five_attempts
Claim	QS-06, BR-11, ADR-008: a failed notification never undoes an activation; messages are retried and marked FAILED after 5 attempts; no restricted data in payloads.
Status	PASSED
Setup	Activation of PLC-9001, then the worker run with a failing channel, with a crash in the middle and with five failures.
Expected	Placement stays ACTIVE; messages PENDING then FAILED at 5; sent messages are not resent after a crash; payload has only placement_id and template.
Observed	As expected.

Limitations	At-least-once towards the channel: a crash between send and commit can repeat one message.
Related	requirements: US-030; qualities: QS-06; decisions: ADR-008
Fixture	DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated
Command	python -m pytest -o addopts="" -v (from the repository root)
Environment	Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema created with create_all (no migrations, BL-13); versions in versions.txt
Independent connections	not applicable
Executor	Sebastián Mercado, 2026-10-11
Executed at	2026-10-11
Trials	1 run; the tests are deterministic
Raw output	docs/evaluation-2/validation/pytest_output.txt
Source digest	f7281a5c6c9e7a50 (command: find src tests config -type f \(-name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' | LC_ALL=C sort | xargs sha256sum | sha256sum | cut -c1-16)

Test	TP-07 / access tests in test_activation_endpoint.py and test_traceability_evidence.py
Claim	QS-01 (partial), QS-02, BR-09, BR-10, ADR-005: relationship-based access, 404 vs 403, audited denials, audience projections.
Status	PASSED
Setup	Seeded data pack; identities from mock_identities.json (student, host-organization contact, coordinator).
Expected	Unrelated user gets 404; related but not allowed gets 403; both audited when the placement exists; non-office audiences never see RESTRICTED evidence references; unknown identity gets 401.
Observed	As expected.
Limitations	The identity mock has no workplace supervisors, so that case is verified at policy level only (EU-05).
Related	requirements: US-027, US-028; qualities: QS-01, QS-02; decisions: ADR-005
Fixture	DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated
Command	python -m pytest -o addopts="" -v (from the repository root)
Environment	Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema created with create_all (no migrations, BL-13); versions in versions.txt

Independent connections	not applicable
Executor	Sebastián Mercado, 2026-10-11
Executed at	2026-10-11
Trials	1 run; the tests are deterministic
Raw output	docs/evaluation-2/validation/pytest_output.txt
Source digest	f7281a5c6c9e7a50 (command: find src tests config -type f \(-name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' | LC_ALL=C sort | xargs sha256sum | sha256sum | cut -c1-16)

Test	TP-08 / tests/architecture/test_module_boundaries.py (5 tests)
Claim	ADR-001, ADR-006, ADR-007: module dependencies follow the written rules; domain code is framework-free; only M-11 imports the mocks; audit is append-only.
Status	PASSED
Setup	Static reading of the import statements of src/horizon.
Expected	No import outside the allowed table; planned modules hold only __init__.py; AuditRecorder exposes only record() and history().
Observed	As expected; mutation M5 (forbidden import) made it fail as intended.
Limitations	Does not see dynamic imports, router wiring or coupling through shared tables.
Related	requirements: -; qualities: -; decisions: ADR-001, ADR-006, ADR-007
Fixture	None: static reading of the source files
Command	python -m pytest -o addopts="" -v (from the repository root)
Environment	Windows 11, Python 3.12.1; no database
Independent connections	not applicable
Executor	Sebastián Mercado, 2026-10-11
Executed at	2026-10-11
Trials	1 run; the tests are deterministic
Raw output	docs/evaluation-2/validation/pytest_output.txt
Source digest	f7281a5c6c9e7a50 (command: find src tests config -type f \(-name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' | LC_ALL=C sort | xargs sha256sum | sha256sum | cut -c1-16)

Test	TP-09 / test_seed_never_modifies_the_raw_files,

	test_seed_loads_every_placement_of_the_pack_once
Claim	EU-03, INV-08: the seed never edits the raw exports and loads every placement once.
Status	PASSED
Setup	Hashes of data/raw before and after the seed; count of placements loaded.
Expected	Raw files unchanged; every placement of placements.csv (four) is present after the seed; the primary key prevents duplicates.
Observed	As expected.
Limitations	The seed is a one-time load, not a migration tool (BL-13).
Related	requirements: -; qualities: -; decisions: ADR-000
Fixture	DH-05 data pack seeded into a throw-away SQLite file for each test session (tests/conftest.py); fixed date 2026-09-24; synthetic PLC-9001 where stated
Command	python -m pytest -o addopts="" -v (from the repository root)
Environment	Windows 11, Python 3.12.1, SQLite 3.43.1 (file in the temp directory), no resource limits, schema created with create_all (no migrations, BL-13); versions in versions.txt
Independent connections	not applicable
Executor	Sebastián Mercado, 2026-10-11
Executed at	2026-10-11
Trials	1 run; the tests are deterministic
Raw output	docs/evaluation-2/validation/pytest_output.txt
Source digest	f7281a5c6c9e7a50 (command: find src tests config -type f \(-name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' | LC_ALL=C sort | xargs sha256sum | sha256sum | cut -c1-16)

Test	TP-10 / docs/evaluation-2/validation/mutation_check.py
Claim	The tests detect real defects in the design mechanisms.
Status	EXECUTED (6 of 6 injected defects detected)
Setup	Each defect injected alone into a copy of the repository; suite run with PYTHONPATH on the copy.
Expected	At least one failing test per defect.
Observed	M1..M6 all DETECTED (mutation_output.txt).
Limitations	Hand-picked defects, not a full mutation-testing tool.
Related	requirements: -; qualities: -; decisions: ADR-001, ADR-005, ADR-007, ADR-008, ADR-010
Fixture	A temporary copy of the repository for each injected defect

Command	python docs/evaluation-2/validation/mutation_check.py . (from the repository root; works on a copy)
Environment	Windows 11, Python 3.12.1, SQLite 3.43.1 (the full suite runs on each copy)
Independent connections	not applicable
Executor	Sebastián Mercado, 2026-10-11
Executed at	2026-10-11
Trials	1 baseline run without defects + 1 run per defect (7 runs)
Raw output	docs/evaluation-2/validation/mutation_output.txt
Source digest	f7281a5c6c9e7a50 (command: find src tests config -type f \(-name '*.py' -o -name '*.json' \) -not -path '*/__pycache__/*' | LC_ALL=C sort | xargs sha256sum | sha256sum | cut -c1-16)

What the tests do not prove:

- The concurrency test interleaves two sessions deterministically on SQLite. It proves the conditional update logic, not behavior under true parallel load on PostgreSQL (TP-14).
- The rollback test proves that nothing is committed when a handler fails; it does not distinguish an explicit rollback from the session being discarded.
- The outbox is at-least-once towards the channel; a crash between send and commit can repeat one message.
- Only the activation slice exists in code (M-03, M-04, M-06, M-07, M-08, M-09, M-11). The four business modules M-01, M-02, M-05 and M-10, the closure gate and the web client are designed, not implemented or tested.
- No performance, security-scanning or accessibility testing has been run.
- The CI workflow (.github/workflows/ci.yml) exists but has not run on a remote, and no pull-request review was exercised: the team worked on one local repository.
- The architecture test reads import statements only. It does not see dynamic imports, imports made inside the routers (api.py, the composition points) or coupling through shared database tables.
- QS-02 with a real workplace-supervisor identity is verified at policy level only, because the identity mock has none (EU-05).

Designed, implemented and verified

Concern	Design	Implementation	Validation
Activation gate, access, audit, outbox	Documented	Implemented	Verified (59 tests, 6/6

Concern	Design	Implementation	Validation
(slice)			mutations)
Concurrency (conditional update)	ADR-010, SEQ-04	Implemented	Verified on SQLite; PostgreSQL not executed
Module boundaries	V-03, V-DEV, dependency rules	Enforced by test	Verified
Opportunities, applications, eligibility	Modules, rules, contracts	Not implemented	Not executed
Hours, evaluations, incidents, closure gate	Modules, transitions, BR-06	Not implemented	Not executed
Web client and UX	V-UX-01, V-UX-02	Not implemented	Not executed
Pilot deployment, real identity, rate limits	V-05 target, T-01, T-10	Not implemented	Not executed

Implementation foundation

The repository is the Horizon starter adapted by ADR-000, with the DH-05 data pack loaded, the module structure of section 6, a CI workflow and one vertical slice working end to end: the activation gate. It exercises four of the five drivers at once (gate, access, audit, external service). Acceptance criteria of the slice, all automated:

- GIVEN placement PLC-2002 with ACADEMIC_APPROVAL (READY-004) and WORK_PLAN (READY-006) pending WHEN the coordinator activates it THEN the activation is refused, the response names READY-004 and READY-006, the status stays PENDING_ACTIVATION AND an audit entry ACTIVATION_REFUSED is written.
- GIVEN placement PLC-2003 with VALID_AGREEMENT expired (READY-007) and INSURANCE_REFERENCE missing (READY-008) WHEN the coordinator activates it THEN it is refused naming both conditions.
- GIVEN a placement whose required conditions are complete and valid and whose organization agreement is ACTIVE WHEN the coordinator activates it THEN the status becomes ACTIVE, an audit entry with previous and new value is written AND a PlacementActivated notification is placed in the outbox.
- GIVEN the Employer Directory answers 503 WHEN the coordinator activates THEN the activation is refused and nothing changes (fail closed).
- GIVEN the notification channel fails WHEN a placement is activated THEN the activation remains committed and the message stays pending for retry.
- GIVEN a student or an external user WHEN they try to activate a placement THEN access is denied and the denial is audited.

The seed script documents every transformation from the raw exports: 'Blocked' is not stored, HU-STU-5006's legacy code SWE becomes BSC-SWE, and INT-APP-1005 says academic approval APPROVED while the checklist of PLC-2003 has no ACADEMIC_APPROVAL row. We report that inconsistency instead of inventing the missing row.

Definition of Done agreed by the team:

- The story's acceptance criteria (Given/When/Then) pass as automated tests, including at least one failure case (missing data, unauthorized user or failed dependency).
- Code merged into main through a pull request reviewed by another team member.
- CI is green: ruff and pytest.
- Access to the new endpoint is declared in the Access module and covered by at least one denial test.
- Every state change writes an audit entry (checked by test).
- Affected documents updated: ADR (new or superseded), module interfaces, traceability.
- No hard-coded program or placement-type rule; rules go to configuration.
- Demonstrable with the DH-05 data pack.

Status for the activation slice: every item is met except the two about the workflow (merge through a reviewed pull request, CI green on a remote). The team works on one local repository and the CI workflow has not run yet; the same commands were run locally and their output is in the validation evidence.

13. Risks, backlog and plan

Architecture risks are about the structure we chose, not the project risks of Evaluation 1. Owners: architecture owner Sebastián Mercado, backend owner Juan Arbeláez, quality owner Cher Jiménez.

ID	Risk	P	I	Mitigation or trigger	Owner role	Status
AR-01	A module imports another module's internals and the boundaries decay.	M	H	Architecture test fails the test suite (and the CI build once CI runs); keep ALLOWED table in sync with the Module Design.	Architecture owner	Mitigated (TP-08)
AR-02	The mock identity header reaches a real pilot as the only authentication.	M	H	No pilot before IdentityAdapter is replaced; decide with IT Services (identity integration is still an open question).	Architecture owner	Open
AR-03	Readiness matrix (CL-020) differs from our union of observed conditions.	H	M	Rules are configuration; replace file and version. Ask the Legal Representative.	Backend owner	Open (EU-02)
AR-04	PostgreSQL behaves differently from SQLite for conditional update or JSON columns.	L	M	Run the suite on PostgreSQL before Evaluation 3 (TP-14).	Quality owner	Open
AR-05	Audit rows altered directly in the database.	L	H	Restrict database permissions in the shared environment; consider hash chaining only if an	Architecture owner	Open

ID	Risk	P	I	Mitigation or trigger	Owner role	Status
				audit requires it.		
AR-06	No rate limiting or abuse protection.	L	L	Accepted at this load (DA-01); revisit with a real pilot.	Backend owner	Accepted
AR-07	Team overload: four business modules (M-01, M-02, M-05, M-10), the closure gate and the web client remain for Evaluation 3.	H	H	Vertical slices in the order BL-01 to BL-05; stop adding scope.	Team	Open
AR-08	CL-018 stays open and the access policy for restricted fields is wrong.	M	H	Deny by default; the policy lives in one module.	Backend owner	Open

ID	Item	Why now	Depends on	Stories	Owner role
BL-01	Applications and eligibility slice (M-02 + M-06 rules + StudentInformationAdapter in use)	Second most important rule set; uses the SIS failure policy (QS-05)	-	US-005, US-007	Backend owner
BL-02	Academic approval and work plan versions; events complete readiness conditions	Feeds the gate with real data instead of seeded rows	BL-01	US-010, US-011	Backend owner
BL-03	Hours in three levels and review	Next highest-risk data (disputes)	BL-02	US-017, US-018, US-019	Backend owner
BL-04	Evaluations, incidents and restricted projections	QS-03; closes the privacy scenarios	BL-03	US-020, US-022	Backend owner
BL-05	Closure gate and CompletionEvidencePort	BR-06	BL-03, BL-04	US-024, US-025	Backend owner
BL-06	Negative tests for invalid rule configuration and directory 404/409 on the activation path; a workplace-supervisor fixture identity for QS-02 end to end	Closes FM-03, FM-04, FM-14, EU-05	-	-	Quality owner
BL-07	Rule publication: who may change a rule version and how it is reviewed	T-06; CL-004, CL-020	Client answer	-	Architecture owner
BL-08	API versioning convention and error format ADR	Contract stability	-	-	Architecture owner
BL-09	Run suite on PostgreSQL with parallel connections	TP-14, AR-04	-	-	Quality owner
BL-10	Web client technology decision	Evaluation 3 UI	BL-01	-	Team
BL-11	Add eventId and schemaVersion to domain events	Event contract before any extraction	-	-	Architecture owner
BL-12	Outbox archive and retention policy	Data lifecycle	-	US-030	Backend owner
BL-13	Versioned schema migrations	create_all cannot change an existing schema	BL-09	-	Backend owner

When	Work	Responsible
By 2026-10-11 (Evaluation 2 delivery)	Submit the Evaluation 2 package (architecture document, ADRs, workbook and repository). Every member rehearses the trace of one requirement in both directions.	Sebastián Mercado, Juan Arbeláez, Cher Jiménez
2026-10-12 to 2026-10-21	Evaluation 3, first iteration: BL-01 applications and eligibility; BL-06 negative tests; BL-08 error format ADR; BL-09 suite on PostgreSQL.	Juan Arbeláez (backend), Cher Jiménez (quality), Sebastián Mercado (BL-08)
2026-10-22 to 2026-10-31	Evaluation 3, second iteration: BL-02 approvals and work plans; BL-03 hours; BL-10 web client decision.	Juan Arbeláez; team for BL-10, led by Sebastián Mercado
2026-11-01 to 2026-11-08	Evaluation 3, third iteration: BL-04 evaluations and incidents; BL-05 closure gate; demo of one placement end to end; update ADRs and views.	Juan Arbeláez, Cher Jiménez, Sebastián Mercado
2026-11-08 (Evaluation 3)	Next architecture review point: does the structure survive execution? Every change goes to a new or superseding ADR.	Sebastián Mercado (architecture owner)
2026-11-29 (Evaluation 4)	Final delivery: remaining backlog items, final evidence and project finished.	Sebastián Mercado, Juan Arbeláez, Cher Jiménez

14. Traceability

Every arrow must be defensible, in both directions. A matrix is the index of the argument, not a substitute for it.

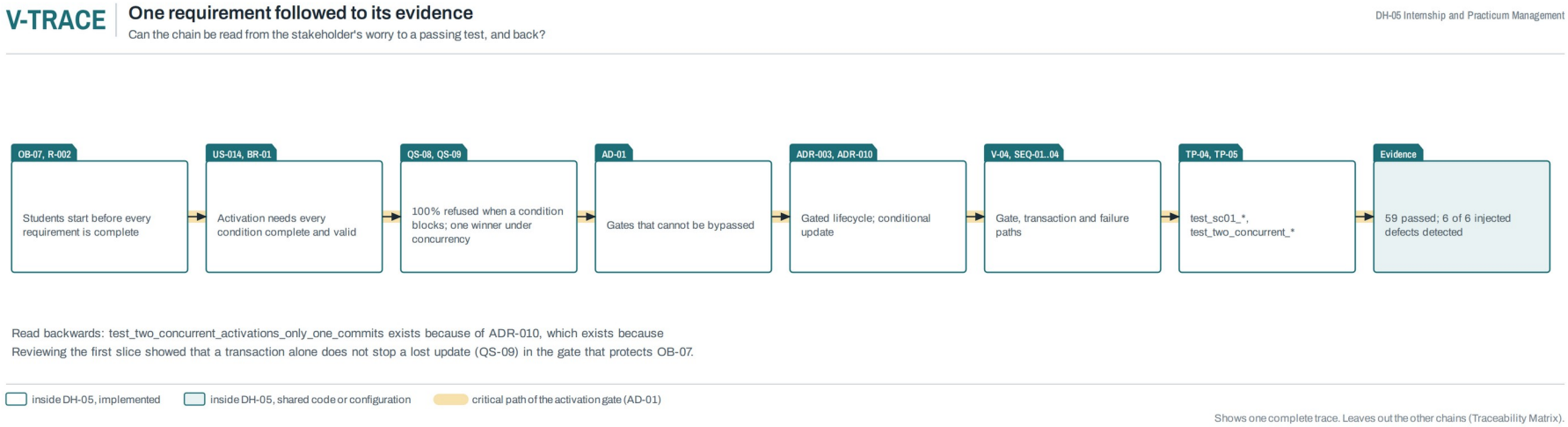


Figure 19. V-TRACE One requirement followed from the stakeholder's worry to the evidence.

Requirement	Rules	Drivers / decisions	Design artifacts	Tests
CAP-04 Readiness gate	BR-01, BR-02, BR-03, BR-04, BR-12	AD-01, AD-05; ADR-003, ADR-004, ADR-007, ADR-010	V-04, V-TXN, SEQ-01..04, lifecycle; ActivatePlacement, Readiness Gate	TP-01, TP-02, TP-04, TP-05
US-027, US-028 Access	BR-09, BR-10	AD-02; ADR-005	V-03, access matrix, threats T-02, T-03, T-08; AccessPolicy	TP-07
CAP-06 History	BR-07, BR-08	AD-03; ADR-006, ADR-010	V-DATA, V-STATUS; Audit Recorder	TP-05, TP-08 (append-only), test_qs04_*
US-030 Notifications	BR-11	AD-05; ADR-008	SEQ-05, V-02; outbox, worker	TP-06
CAP-02 Eligibility	BR-08	AD-04, AD-05; ADR-004, ADR-007	M-02, M-06, StudentInformation Port	TP-03 (adapter); TP-12 NOT EXECUTED
CAP-03 Three approvals	BR-05	AD-03; ADR-006	M-02, V-DOM	TP-12 NOT EXECUTED
CAP-05 Verified hours	BR-06	AD-03, AD-04; ADR-004, ADR-006	M-05, V-DOM	TP-12 NOT EXECUTED
CAP-06 Closure	BR-06	AD-01; ADR-003	lifecycle, closure gate	TP-12 NOT EXECUTED
CAP-01 Publication	-	ADR-001, ADR-006	M-01	TP-12 NOT EXECUTED

Driver	Scenarios	Evidence	Decisions	Modules / patterns	Verification
AD-01	QS-08, QS-09	OB-07, OB-11, ASM-006, R-002, R-007	ADR-001, ADR-002, ADR-003, ADR-010	M-03, M-04, M-05; Facade (P-06)	SC-05, SC-12; test_activation_endpoint.py (test_sc01_*, test_sc02_*, test_directory_down_*); test_concurrency_and_atomicity.py
AD-02	QS-01, QS-02, QS-03	CL-009, CL-018, ASM-007, R-003, R-004	ADR-005	M-07 (used by all); -	SC-07, SC-08, SC-09; test_student_cannot_activate_*, test_unrelated_user_*, test_host_organization_sees_*, test_qs02_*
AD-03	QS-04, QS-09	OB-12, F-02, CL-019	ADR-002, ADR-006, ADR-010	M-08 (used by all); -	SC-10; audit assertions in test_activation_endpoint.py; test_one_correlation_id_*; test_qs04_*
AD-04	QS-07	F-24, CL-004, CL-020, ASM-004, R-001, EU-01, EU-02	ADR-004	M-06; Strategy (P-01), Factory Method (P-02)	SC-03, SC-05; tests/unit/test_readiness_rules.py::test_rule_change_is_configuration_only
AD-05	QS-05, QS-06, QS-08	F-17, F-18, ASM-002, ASM-003, ASM-009, R-005	ADR-007, ADR-008	M-09, M-11; Adapter (P-03), Observer (P-04)	SC-11; tests/unit/test_reference_and_adapters.py, tests/integration/test_outbox.py, test_qs06_*

Story	Module	Driver	Decision	Component / pattern	Success
US-001	M-01	-	ADR-001	OpportunityService	SC-01
US-002	M-01	AD-04	ADR-004	OpportunityService	SC-01
US-003	M-01	AD-03	ADR-006	OpportunityService, audit	SC-01
US-004	M-01	AD-02	ADR-005	Search + projection	SC-02
US-005	M-02	AD-05	ADR-007	ApplicationService	SC-02
US-006	M-02	-	ADR-001	ApplicationService	SC-02
US-007	M-02, M-06	AD-04, AD-05	ADR-004, ADR-007	Strategy (P-01), Adapter (P-03)	SC-03
US-008	M-02	AD-02	ADR-005	Projection	SC-02
US-009	M-02	AD-03	ADR-006	ApplicationService, events	SC-04
US-010	M-03	AD-03	ADR-006	WorkPlan versions	SC-04
US-011	M-02	AD-01	ADR-003	Three decision records	SC-04
US-012	M-04, M-06	AD-04	ADR-004	Readiness checklist, Strategy	SC-05
US-013	M-04	AD-03	ADR-006	Evidence references	SC-05
US-014	M-03, M-04	AD-01, AD-05	ADR-003, ADR-007	ActivationGate, EmployerDirectoryPort	SC-05
US-015	M-03	AD-02, AD-03	ADR-005, ADR-006	Supervisor assignment (relationships)	SC-05
US-016	M-03	AD-01	ADR-003	Lifecycle transition table	SC-05
US-017	M-05	AD-03	ADR-006	HourEntry	SC-06
US-018	M-05	AD-03	ADR-006	Hour review records	SC-06
US-019	M-05	AD-03	ADR-006	Dispute resolution	SC-06
US-020	M-05, M-07	AD-02	ADR-005	Incident visibility	SC-09
US-021	M-03	AD-03	ADR-006	WorkPlan versions	SC-10
US-022	M-05, M-07	AD-02	ADR-005	Evaluation projections	SC-07
US-023	M-05	-	ADR-001	Reflection record	SC-07
US-024	M-05, M-06	AD-01, AD-04	ADR-003, ADR-004	Closure gate, Strategy	SC-12
US-025	M-03	AD-01	ADR-003	Lifecycle transition table	SC-12
US-026	M-08	AD-03	ADR-006	Append-only audit	SC-10
US-027	M-07, M-11	AD-02, AD-05	ADR-005, ADR-007	IdentityAdapter	SC-11
US-028	M-07	AD-02	ADR-005	Access policy + projections	SC-08, SC-09
US-029	M-10	-	ADR-002	Read-only queries	SC-02
US-030	M-09	AD-05	ADR-008	Observer (P-04), outbox	SC-11

Reverse traces

Start from the worker (C-WORKER). It exists because a failed notification must never cancel an activation (US-030, BR-11, QR-04, QS-06). ADR-008 chose a transactional outbox. The worker reads PENDING rows from C-DB, calls the channel and commits per message without ever touching a placement. It appears in V-02, V-05 and SEQ-05, and TP-06 challenges it. A worker without the outbox table or without the crash test would be a broken reverse trace.

Start from the conditional update in SqlPlacementRepository.save. It exists because two coordinators can act on the same placement (QS-09, BR-12). ADR-010 chose it over row locks. It is not an authorization mechanism and does not replace the gate; it only guarantees one winner. SEQ-04 shows it and test_two_concurrent_activations_only_one_commits verifies it on SQLite.

15. What is not decided yet

Eleven clarifications of Evaluation 1 are still open; seven affect the architecture. Each unknown sits behind configuration or behind one module, so the client's answer changes one place.

Clarification	Still unknown	How the design stays safe meanwhile	Decision
CL-008	Who may authorize an activation with a missing condition	The gate supports an exception only as a separate, audited command; not implemented until the role is confirmed.	ADR-003
CL-017	Who resolves disputed hours	The resolving role is a configuration value (default: experiential-learning coordinator, US-019).	ADR-004
CL-018	Who may read restricted comments and sensitive incidents	Deny by default except the office, the faculty supervisor of the placement and the author of the content (a workplace supervisor reads their own comments); policy lives only in M-07.	ADR-005
CL-020	Mandatory readiness conditions per placement type	Initial configuration from observed data (EU-02), versioned, replaceable without code.	ADR-004
CL-004	Exact eligibility rules per program	Rule types derived from eligibility_reviews.csv; parameters per program in configuration.	ADR-004
CL-016	Whether the reflection is mandatory to close	Completion requirement configurable per program.	ADR-004
CL-001	Placement types in the first release (ten defined, five used by opportunities, three by placements)	Placement types are reference data; rules are configured per type; the demo uses the types present in the scenarios.	ADR-004

16. How this evaluation builds on Evaluation 1

Nothing from Evaluation 1 was discarded or rewritten. Its artifacts remain the reference for the problem, and this document uses them as inputs: the quality requirements became the measurable scenarios of section 3, the risks and clarifications became drivers and open questions (sections 4 and 15), the process stages became the module boundaries (section 6), and the stories and success criteria became the traceability and the tests (sections 12 and 14). Where the data pack or the starter kit added detail, we recorded it as an evidence update in section 1 instead of editing the earlier delivery.

17. Evidence index

For every important claim: where to look and how strong the evidence is. The repository README is the entry point and lists the same index with paths.

Claim	Where to look	Status
The structure is a modular monolith with 11 modules	Architecture Document s.6-7; ADR-001; V-02, V-03	DESIGNED + boundaries VERIFIED (TP-08)
Dependencies between modules follow the written rules	tests/architecture/test_module_ boundaries.py	VERIFIED (5 tests)
The activation gate cannot be bypassed and names every blocking condition	tests/integration/test_activation_endpoint. py; tests/unit/test_placement_lifecycle.py	VERIFIED
Two concurrent activations produce one success	tests/integration/test_concurrency_and_ atomicity.py	VERIFIED on SQLite (interleaving); parallel PostgreSQL NOT EXECUTED
A failure in the middle of a command leaves no partial data	test_failure_in_the_middle_rolls_ everything_back	VERIFIED
A notification failure never undoes an activation	tests/integration/test_outbox.py; test_ qs06_*	VERIFIED
Restricted evidence is hidden from non-office audiences	test_host_organization_sees_readiness_ without_restricted_evidence	VERIFIED (evidence references only)
Evaluation and incident projections hide restricted fields	Access Policy sheet; ADR-005	DESIGNED (module M-05 not built)
Rules can change without code changes	tests/unit/test_readiness_rules.py::test_ rule_change_is_configuration_only; config/ rules/readiness.json	VERIFIED for readiness rules; eligibility rules DESIGNED
The tests detect real defects	docs/evaluation-2/validation/mutation_ output.txt	EXECUTED: 6 of 6 injected defects detected
The suite passes and the code is lint-clean	docs/evaluation-2/validation/pytest_	EXECUTED on the delivery

Claim	Where to look	Status
	output.txt, ruff_output.txt, versions.txt	date (see Validation evidence)
The data pack loads and validates	scripts/validate_data.py; data/generated/ validation_report.json	EXECUTED: 120 records accepted, 0 errors, 25 warnings
Every transformation of the raw exports is recorded	data/generated/transformation_log.md (written by scripts/seed_domain.py)	IMPLEMENTED; raw files unchanged VERIFIED
Traceability in both directions	Traceability Matrix sheet; Architecture Document s.14	Reviewed by the authors before delivery; not automated in the repository
Closure gate, eligibility, hours, evaluations	Module Design; Backlog BL-01..BL-05	DESIGNED only
Behavior on PostgreSQL, load, authentication	Test Plan TP-13..TP-15	NOT EXECUTED

18. Glossary

Term	Meaning in this document
Architecture driver	A requirement, scenario or constraint whose change would change the structure.
ADR	Architecture Decision Record: context, decision, alternatives, consequences, verification, revisit triggers.
Audience projection	The version of a resource that one kind of user may see, with forbidden fields removed.
Conditional update	A write that succeeds only if the stored value still equals the value that was read (UPDATE … WHERE status = the status read). Also called compare-and-set or optimistic concurrency.
Composition point	The place where modules, adapters and mocks are wired together; in this project the routers (modules/*/api.py).
Shared kernel	Small shared code that every module may use: the ports and the domain events (horizon/ shared).
Fail closed / fail soft	Refuse the operation when a dependency fails / keep the request and finish it later.
Modular monolith	One deployable application with explicit internal module boundaries.
Port / adapter	An interface owned by the core / the code that implements it for one external system.
Transactional outbox	Messages written in the same transaction as the decision and delivered later.
Readiness gate	The evaluation of every required condition before activation.
Rule version	The label of the rule configuration used for a decision.
Invariant	A property that every committed state must keep.

Term	Meaning in this document
Evidence	The recorded result of something we actually ran; not a diagram or an expected result.

19. References

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